

IS THERE A PLATONIC TREE?

CALIBRATING LATENT HYPERBOLICITY

IN FOUNDATION MODELS

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ABSTRACT

Hyperbolic methods for representation learning rest on a premise we call latent hyperbolicity: standard models are already tree-like, because their class geometry scores a low Gromov δ . We show that a low raw reading is what high dimension, an anisotropic spectrum and a maximum over sampled quadruples produce on their own, and we build an instrument that reads every score as an excess over a random cloud of the same shape, ranks it against 200 matched replicates, and adds a depth test whose false-alarm rate and power are measured on real clouds. Across 12 vision backbones, 6 datasets and 16 text models, the premise does not survive: on image features the excess sits within null noise in most cells (model-dataset pairs). Class centroids do carry clustered structure in 49 of 72 cells, but a star already produces it; hierarchy above the superclasses is certified in only 4 of 12 ImageNet backbones, by a test that never fires on randomized hubs, and a backbone trained in hyperbolic space shows the same clustering, no detected depth, and embeddings that never leave the near-flat regime. The trees are moderately shared: the naive comparison manufactures an island, and once the clustering cut is controlled every recipe recovers the human taxonomy partially, with the self-supervised tree living in angles rather than distances. Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry.

1 INTRODUCTION

Semantic knowledge is hierarchical, and hyperbolic space is the natural geometric home for hierarchies (Nickel & Kiela, 2017; Sarkar, 2011; Ganea et al., 2018; Chami et al., 2019). A line of work therefore imposes it on representations at training time, from vision to vision-language and language (Khrulkov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024; Desai et al., 2023; Pal et al., 2025; Chen et al., 2022; He et al., 2025; Sinha et al., 2024). These methods cite a premise, which we call *latent hyperbolicity*: standard models already organize concepts in a tree-like metric structure. The evidence is a single number, the Gromov δ of their features: δ is zero for a metric tree and grows as a metric departs from one, and the measured values are low, on CNN and ViT features (Khrulkov et al., 2020; Bdeir et al., 2024), on token embeddings (Yang et al., 2025) and on word vectors (Tifrea et al., 2019). Those readings are taken at the sample level, on individual images or tokens rather than on classes, and they are never calibrated: a value is called low on its own, without asking what a structureless cloud of the same dimension and spectrum would score. Is the premise right, and what does it license? Figure 1 shows why a low reading is not enough: a structureless cloud, a star of clusters without depth and a tree with depth all read the same.

figures/fig1_concept.pdf (conceptual figure, drawn by the authors; same-size placeholder)

Figure 1: **Plato’s cave: a random cloud, a star and a tree cast the same shadow.** An observer who sees only shadows cannot tell which object cast them, and a raw δ is such a shadow: points with no structure, clusters around a hub, and a nested hierarchy all read as low δ . The instrument reads the objects instead of the shadow: no structure beyond the null in the cloud, clusters without depth in the star, clusters with depth in the tree.

Why is the reading low? Gromov δ assigns zero to trees (Gromov, 1987), but three artifacts push it down without any hierarchy. High-dimensional clouds concentrate toward equidistance (Beyer et al., 1999; Aggarwal et al., 2001), which is a degenerate star tree: we call this the *dimension confound*, the geometric counterpart of the width confounder of Gröger et al. (2026). Anisotropic spectra mimic low-dimensional behavior: the *spectrum confound*. And the supremum over sampled quadruples is a one-quadruple extreme that does not converge (Fournier et al., 2015): the *statistic confound*. The three are visible in real models: GPT-2 M sits at its matched null despite a low raw reading, and classic sentence embedders share their raw reading with vision-contrastive models while carrying none of their structure.

We build the instrument that reads δ correctly. It calibrates the estimator on reference geometries, expresses every reading as an *excess* over a null cloud matched in dimension and covariance spectrum, ranks that excess against matched replicates, and adds a *depth test* whose false-alarm rate and power are measured on real clouds. With it we ask of geometry what convergence work asks of similarity (Huh et al., 2024; Gröger et al., 2026): what survives calibration, and is it shared? They calibrate similarity across models; we calibrate geometry within a model, and comparing trees across models needs a calibration of its own.

The answer has three parts. On image features the excess sits within null noise in most cells and is a fraction of the null where it survives, so the premise does not survive calibration as stated. On class centroids, clustered structure is genuine in 49 of 72 cells, but a star already produces it. Hierarchy above the superclasses is certified in 4 of 12 ImageNet backbones by a test that never fires on real clouds with randomized hubs, and a backbone trained in hyperbolic space shows the same clustering and no detected depth. The trees are moderately shared: the naive comparison manufactures an island, and once the clustering cut is controlled a moderate gap remains, every recipe recovers the human taxonomy partially, more so with supervision, and the self-supervised structure lives in angles.

Our contributions are:

1. **The instrument.** Estimator calibration, spectrum-matched nulls read as corrected ranks, and a depth test with measured false-alarm rate and power.
2. **The census.** Clustered structure in most models, hierarchy certified in a few, and a hyperbolic backbone as the control for imposing the geometry.
3. **The map of trees.** The island is an artifact of the cut; sharing is graded and supervision-dependent, and the self-supervised tree is angular.
4. **Consequences for practice.** What to measure before imposing curvature, and which zero-cost readout collects what is there.

2 RELATED WORK

Reading δ against a matched reference has precedent in network science, where the δ of real graphs is compared with that of random graphs of comparable size (Narayan & Sanjeev, 2011; Kennedy et al.,

2013; Adcock et al., 2013); we bring the idea to embedding clouds, where the reference must match dimension and spectrum. The analyses of learned representations characterize Euclidean structure and do not test for tree-likeness: representation similarity, intrinsic dimension and neural collapse (Kornblith et al., 2019; Ansuini et al., 2019; Pope et al., 2021; Pappan et al., 2020). Closest to us is the work on representational convergence: the Platonic Representation Hypothesis (Huh et al., 2024) argues that models converge toward shared representations, and its cross-modal convergence has been stress-tested at scale (Koepeke et al., 2026); Gröger et al. (2026) show that permutation-calibrated similarity removes most of that convergence while local neighborhood structure survives, and Park et al. (2024) read concept hierarchy in language models from the orthogonality of concept directions. We are the geometric complement of this program: the same calibration applied within a model instead of across models, with a depth test measured on real clouds, a hyperbolic backbone as the control for imposing the geometry, and a calibration of the tree-to-tree comparison itself.

3 METHODOLOGY

3.1 BACKGROUND: GROMOV δ

Tree-likeness is measured on quadruples. Take any four points w, x, y, z of a metric space with distance d . They can be paired in three ways, and each pairing has a sum of two distances:

$$d(w, x) + d(y, z), \quad d(w, y) + d(x, z), \quad d(w, z) + d(x, y). \quad (1)$$

Sort the three sums as $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$. In a tree the two largest sums are always equal, because the two pairings that cross the trunk traverse the same path, so the four-point defect $(S_{\max} - S_{\text{mid}})/2$ is zero; the further a metric is from a tree, the larger the defect can become. Gromov δ is the worst case, the supremum of the defect over all quadruples (Gromov, 1987). A metric tree has $\delta = 0$. Euclidean space has no bound at all: enlarging a square enlarges its defect in proportion. Hyperbolic space sits in between, with a bounded defect whose value for the hyperbolic plane is known, $\ln 2$ under this definition (Nica & Špakula, 2016).

Definition 1 (four-point defect). For a quadruple (w, x, y, z) with the pairing sums of Eq. 1 sorted as $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$, the four-point defect is $\delta(w, x, y, z) = \frac{1}{2}(S_{\max} - S_{\text{mid}})$, and the Gromov δ of a metric space is the supremum of the defect over its quadruples.

3.2 ESTIMATION AND NORMALIZATION

Three steps turn δ into a number that can be compared across models. First, a thousand centroids have far more quadruples than can be enumerated, so the defect is computed on half a million random quadruples. Second, the largest defect in that sample is a poor estimate of the supremum: it belongs to a single extreme quadruple and keeps growing as more quadruples are drawn (Fournier et al., 2015). We therefore read the 99.9th percentile of the sampled defects, $\hat{\delta}_{99.9}$, the value that 99.9 per cent of them fall below: it keeps the tail of the defects, as the supremum does, but it is set by hundreds of quadruples rather than by one. It also does not move with the sample size: across quadruple budgets the ImageNet reading drifts by less than half its own spread (Table 2). Third, the defect grows with the size of the point set, so we divide by the diameter, the largest pairwise distance:

$$\delta_{\text{norm}} = \frac{\hat{\delta}_{99.9}}{\text{diam}}. \quad (2)$$

On this scale a tree reads zero, and a hyperbolic region of radius four or more reads below a sphere (Table 3).

Definition 2 (reading of record). The reading of record of a cloud is δ_{norm} of Eq. 2, with $\hat{\delta}_{99.9}$ computed over half a million sampled quadruples per seed and averaged over ten seeds.

Proposition 1 (range bound and dimension confound). *Let $d_{\min}$ and $d_{\max}$ be the smallest and the largest pairwise distance of a finite metric space. (a) For any quadruple the four-point defect is at most $d_{\max} - d_{\min}$, hence $\delta_{\text{norm}} \leq 1 - d_{\min}/d_{\max}$. (b) For an iid Gaussian cloud of n points in dimension d , $d_{\min}/d_{\max} \rightarrow 1$ as d grows (Beyer et al., 1999; Aggarwal et al., 2001), so $\delta_{\text{norm}} \rightarrow 0$ without any structure.*

A structureless cloud reads low. Proposition 1, proved in Appendix A, says that a high-dimensional cloud with no structure reads as tree-like. An anisotropic spectrum moves the bound

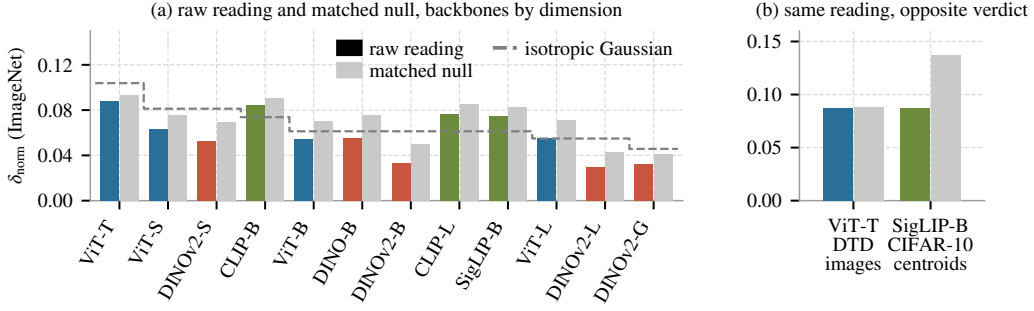


Figure 2: **The raw reading cannot be read on its own.** (a) Raw reading and matched null per ImageNet backbone, ordered by dimension, with the isotropic Gaussian reference dashed; the gap between the two bars is the excess. (b) Two cells with the same raw reading and opposite verdicts, ViT-T on DTD images and SigLIP-B on CIFAR-10 centroids.

through the effective dimension, and a supremum over sampled quadruples grows with the sample. The calibration on reference geometries confirms this: the estimator recovers the tree zero and the $\ln 2$ of the hyperbolic plane, spheres stay high, and iid Gaussians drift down with the dimension (Table 3).

3.3 SPECTRUM-MATCHED NULL, EXCESS AND GENUINENESS

Definition 3 (Haar null). Let $X = U\Sigma V^\top$ be the singular value decomposition of the centered cloud. A null replicate is $X' = Q\Sigma V^\top$ with Q Haar-distributed on the orthogonal group, so X' has the singular values of X and structureless coefficients.

The null is a cloud with the real shape and no structure. Gaussian coefficients reproduce the spectrum only in expectation, fall short of it when n is small relative to d , and inflate the excess of low-rank clouds. The Haar construction reproduces the sample spectrum exactly, so it is the record. The Gaussian reading is reported beside it (Table 4).

Definition 4 (excess). With K null replicates $X'_1, \dots, X'_K$ of Definition 3, the excess of a cell is

$$e(X) = \delta_{\text{norm}}(X) - \frac{1}{K} \sum_{k=1}^K \delta_{\text{norm}}(X'_k). \quad (3)$$

The rank of the reading is the evidence. The excess is a size. The evidence is the rank of the reading among its replicates and the left-tail p that the rank gives,

$$r = \#\{k : \delta_{\text{norm}}(X'_k) > \delta_{\text{norm}}(X)\}, \quad p = \frac{1 + K - r}{K + 1}, \quad K = 200. \quad (4)$$

Rank-based calibration gives the score its finite-sample validity, and a z -score is kept only as a secondary summary.

Definition 5 (genuine). Order the m left-tail p -values of a census as $p_{(1)} \leq \dots \leq p_{(m)}$. A cell is genuine when its p is at or below the Benjamini–Hochberg threshold at level five per cent,

$$p \leq p_{(i^*)}, \quad i^* = \max\{i : p_{(i)} \leq 0.05 i/m\}, \quad (5)$$

and the word is used in this sense only.

A negative genuine excess means clustered, not tree-like. A negative genuine excess certifies clustered structure. The word tree-like is reserved for the depth test of Section 3.4. The null moves with the dimension, so the reading is the gap between the raw bar and the null, and two cells with the same raw reading can receive opposite verdicts (Figure 2).

3.4 DEPTH TEST

Definition 6 (hub null and excess B). Given a frame that assigns each centroid x_i to a cluster $k(i)$ with hub $h_{k(i)}$, the cluster mean, write $x_i = h_{k(i)} + o_i$. The hub null keeps every offset o_i and

replaces the hubs by a Haar resample with their spectrum, $x'_i = h'_{k(i)} + o_i$, and the excess of Eq. 3 against this null, e_B , measures hierarchy above the clusters of the frame only.

Definition 7 (matched star). The matched star of a cloud keeps the real clusters, each as a Haar sample with the cluster’s own covariance, and places their hubs as Gaussian draws at the real hub radius, so it carries the clustering of the real cloud and no hierarchy above it. Its spectrum-matched-hub variant draws the hubs as a Haar resample of the real hubs instead.

The depth test compares the real cloud with its matched star. With e_B read on the real cloud and on its matched star, the depth statistic and its standardized form are

$$\text{depth} = e_B(X) - e_B(X_{\text{star}}), \quad z = \text{depth} / \text{s.d.}, \quad (6)$$

The spread is taken over star seeds and null replicates, and a cloud is certified as more hierarchical than a star when $z \leq -2$. The test is needed because a star with no hierarchy passes the census of Definition 5. The frame is the WordNet cut on ImageNet and the coarse labels on CIFAR-100, and the false-alarm rate and power are measured on real clouds in Section 5.3.

3.5 COMPARING TREES ACROSS MODELS

Trees are compared under a configuration chosen before the comparison. Each model’s hierarchy is a dendrogram over the shared class centroids. Two hierarchies are compared by the adjusted Rand index between their cut partitions, by the correlation of their cophenetic distances, and by their agreement on which pair of a random class triplet merges first. The configuration of metric and linkage is selected per dataset by a criterion fixed without reference to the agreements: the highest cophenetic fidelity among configurations whose reference cuts are non-degenerate for every model. Local agreement is read with the permutation calibration of Gröger et al. (2026), on mutual-kNN agreement and on centroid CKA, with permutations of the class correspondence as the null.

3.6 PROJECTION AND TASKS

Three readouts are compared on the intact representation. A parameter-free map centers the features, scales the 95th-percentile norm to radius $1/\sqrt{2}$ and preserves directions. Euclidean, cosine and Poincaré readouts are then compared on the same representation. Nearest-centroid classification and 5-way 5-shot episodes score the three readouts on every cell of the task grid.

4 EXPERIMENTAL SETUP

The census has twelve backbones, six class sets and sixteen text models. The vision backbones are four supervised ViTs trained on ImageNet-21k, DINO-B, four DINOv2 sizes, CLIP-B, CLIP-L and SigLIP-B. The class sets are ImageNet and CIFAR-100, which carry a real hierarchy, CIFAR-10 and DTD, and the flat FMNIST and MNIST. The text models read the thousand ImageNet class names: four GPT-2 sizes, three Pythia sizes, two OLMo sizes and seven sentence embedders, which also read DBpedia Classes and HierarCaps. Table 5 lists the panel.

Four controls test the alternatives. Two self-supervised ResNet-50 backbones read the vision census. Two ViT-B trained on ImageNet-1k leaf labels, DeiT-B and the augreg ViT-B, test whether a hierarchical label set explains the depth. MERU and its Euclidean CLIP twin, in three sizes, test whether training in hyperbolic space creates depth. The census ViT-B fine-tuned with and without a hierarchical objective is a trained control.

The premise is read on images and hierarchy on centroids. A cell is one pair of model and class set. The sample-level reading uses about a thousand stratified training images per cell on CIFAR-100 and DTD. Class centroids average a hundred cached training images per class on ImageNet and the full cached split elsewhere. Causal language models give the last-token state of “a photo of a {class}”, extracted one prompt at a time because batching alters GPT-2’s hidden states, and embedders use their recommended pooling (Table 6).

Every reading is calibrated against two hundred replicates. Each cell is read against two hundred Haar replicates, with half a million sampled quadruples per seed and ten seeds. Benjamini-Hochberg runs over the vision census and, separately, over the text census. The depth test uses

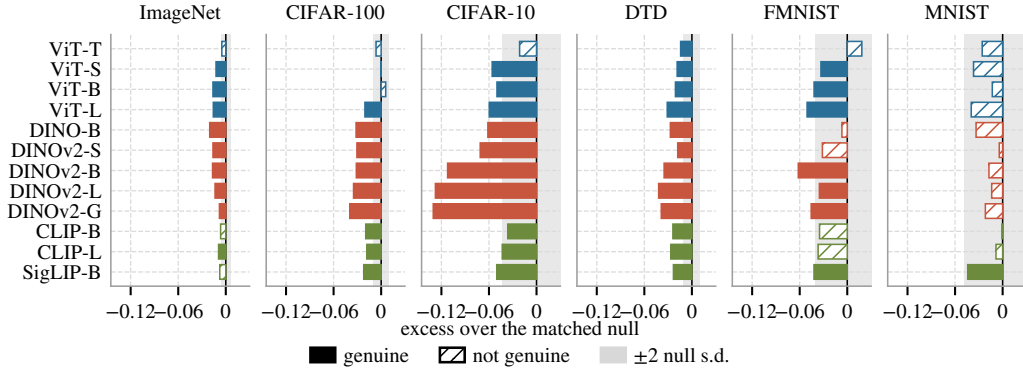


Figure 3: **Most cells show clustered structure.** Excess of the reading of record over its matched null, one panel per dataset and one bar per backbone, filled when genuine and hatched when not, with the band of two null standard deviations around zero.

model	family	$\hat{\delta}_{99.9}$ IN	excess IN	$r/200$ (p)	exc. C100	exc. C10	ρ_{WN}
ViT-T	sup.	0.088	-0.005°	169 (.159)	-0.006°	-0.021°	+0.52
ViT-S	sup.	0.063	-0.012	200 (.005)	-0.000°	-0.056	+0.50
ViT-B	sup.	0.054	-0.016	200 (.005)	$+0.006^\circ$	-0.050	+0.49
ViT-L	sup.	0.055	-0.016	200 (.005)	-0.020	-0.060	+0.53
DINO-B	SSL	0.055	-0.020	200 (.005)	-0.032	-0.061	+0.36
DINOv2-S	SSL	0.053	-0.016	200 (.005)	-0.031	-0.071	+0.22
DINOv2-B	SSL	0.034	-0.017	200 (.005)	-0.031	-0.112	+0.18
DINOv2-L	SSL	0.030	-0.014	200 (.005)	-0.035	-0.128	+0.18
DINOv2-G	SSL	0.033	-0.008	200 (.005)	-0.040	-0.130	+0.20
CLIP-B	contr.	0.084	-0.007°	172 (.144)	-0.019	-0.036	+0.57
CLIP-L	contr.	0.076	-0.009	197 (.020)	-0.018	-0.043	+0.59
SigLIP-B	contr.	0.075	-0.008°	189 (.060)	-0.022	-0.050	+0.58

Table 1: **Clustered structure is the rule at the class level.** One row per backbone: raw ImageNet reading $\hat{\delta}_{99.9}$, its excess over the mean of 200 Haar replicates, rank $r/200$ with left-tail p , genuine = Benjamini–Hochberg $p \leq 0.05$ over 72 cells ($^\circ$: not genuine; bold: less clustered than the null), the CIFAR-100 and CIFAR-10 excess, and ρ_{WN} , the Spearman correlation with WordNet distances.

the WordNet cut into thirty superclasses on ImageNet, the twenty coarse labels on CIFAR-100, a balanced frame as a control, and ten star seeds.

5 RESULTS

5.1 THE PREMISE AT THE SAMPLE LEVEL

The premise fails where it is read. On per-image features the raw reading lies in the low band the literature reports. Calibrated against the matched null, it is not genuine in most cells. Where it is genuine, the excess removes at most two fifths of the null reading, most for DINOv2-G on CIFAR-100. The genuine cells belong mostly to the DINOv2 family, whose structure is the angular tree of Section 5.4. Table 7 gives the reading per cell.

5.2 CLUSTERED STRUCTURE AT THE CLASS LEVEL

Most cells show clustered structure. On class centroids the excess is negative in almost every cell. It is genuine in 49 of 72 cells. On ImageNet and CIFAR-100, the two sets with a real hierarchy, the count is 18 of 24. Figure 3 shows the census and Table 1 gives it row by row.

The count survives resampling. The census null accounts for replicate noise but not for image resampling or the estimator’s seeds. Folding all three into one spread, or repeating the census on

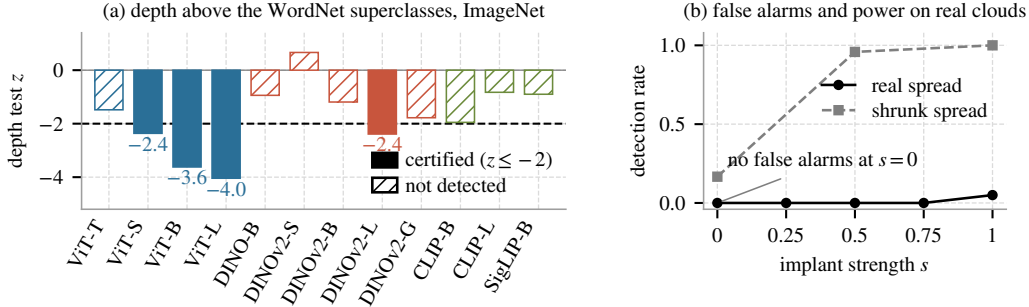


Figure 4: **Depth above the superclasses is certified in 4 of 12 ImageNet backbones, and the test never fires on randomized hubs.** (a) Depth z per backbone under the matched star, filled and numbered when certified, hatched otherwise, with the certification line at $z = -2$. (b) Detection rate against implant strength on the real clouds with the real within-cluster spread (solid) and with it shrunk into the validated range (dashed).

resampled centroid sets, leaves most of the count in place. The cells that move are on the transfer sets. Table 2 gives the counts.

No family owns the structure. Every family contributes genuine cells. The smallest backbone is genuine on one dataset only, and the flat datasets sit near zero. A class-count control shows that the reading follows the hierarchy of the labels rather than their number. Two self-supervised ResNets read like the ViTs. Tables 4 and 2 give the details.

A star of clusters already passes the census. A star of clusters has structure beyond second moments, and a synthetic star with no hierarchy scores an excess as deep as any real cell. A negative excess therefore says clustered, not deep. Table 3 gives the star’s excess.

Neural collapse is the flat limit, not what the census sees. Neural collapse predicts that supervised class means converge to a simplex frame (Papayan et al., 2020), a star with no hierarchy. The arrangement we certify is not that frame: class means cluster around superclass hubs, and the same structure appears without labels and in language models. Scale deepens the excess for DINOv2 only, on CIFAR-10 and weakly on CIFAR-100. Training moves the raw reading within an architecture, up after fine-tuning without class hierarchy and down across depth. Table 2 gives the training rows.

5.3 DEPTH ABOVE THE LABELLED CLUSTERS

Four backbones are certified on ImageNet. The real cloud is more hierarchical than its matched star in 4 of 12 backbones, ViT-S, ViT-B, ViT-L and DINOv2-L, and never less. Under the spectrum-matched-hub star the certified set is the same four. The other eight are not detected, which is not the same as not hierarchical. Figure 4a shows the z per backbone and Table 8 gives both stars.

The test never fires on randomized hubs. With the hub arrangement of every real cloud randomized, the test fires in 0 of 60 runs, and the four certified backbones with random hubs never fire either. Their verdict therefore comes from the real hub arrangement. Table 9 gives the runs.

The test is weak at ImageNet’s noise level. A clean implanted two-level tree is detected in almost no run, because the within-cluster spread of the real clouds exceeds their between-hub spread, outside the range the synthetic sweep covered. Once the offsets are shrunk into that range every backbone is detected, but two contrastive backbones then fire at zero strength, so the shrunk variant only measures power. Figure 4b shows the detection rates and Table 9 gives the power.

The certified set depends on the frame. CIFAR-100 has a hundred centroids in twenty clusters, outside the validated regime, so its readings carry no certification. An earlier isotropic star produced false alarms and is withdrawn. Table 8 gives the frames.

Leaf labels can produce the depth but do not guarantee it. The augreg ViT-B trained on ImageNet-1k leaf labels is certified and aligned with WordNet like the ImageNet-21k ViTs. DeiT-B, trained on the same labels with another recipe, is neither certified nor aligned, yet it recovers the

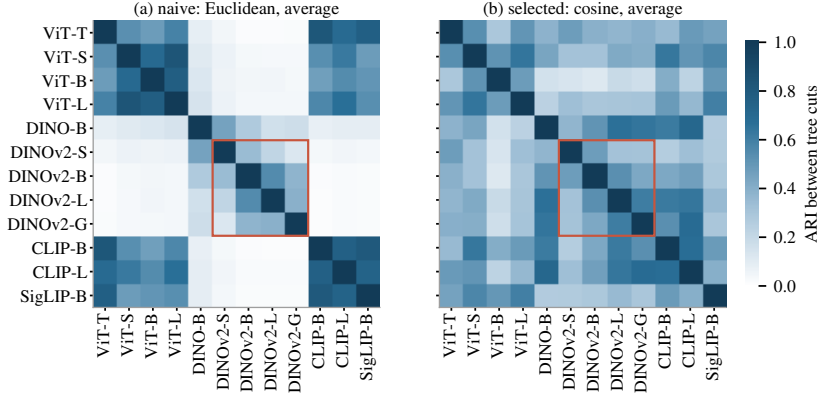


Figure 5: **The island is a chaining artifact, and a moderate gap remains.** Pairwise ARI between the dendrogram cuts of the 12 vision models on ImageNet under the naive configuration (a), where the DINOv2 block in the rectangle agrees with the rest at 0.03, and under the selected configuration (b), where it agrees at 0.38.

CIFAR-100 superclasses. A hierarchical label set is therefore not the explanation of the certified ViTs’ depth, and alignment with the taxonomy is recipe-dependent. Table 8 gives the two models.

Imposing the geometry does not create detected depth. MERU embeds images on the Lorentz hyperboloid with an entailment objective, and its Euclidean twin shares backbone, data and recipe (Desai et al., 2023). The two read the same excess, and neither is detected as more hierarchical than a matched star. MERU’s embeddings never leave the near-flat regime, where Lorentz and Euclidean distances almost coincide, so the control tests the objective rather than a curved geometry. Table 8 gives the readings and the radii.

5.4 WHOSE TREE

The naive map manufactures an island. Under Euclidean distances with average linkage the DINOv2 family agrees with the supervised and contrastive block far less than the block agrees with itself. The cause is chaining: DINOv2’s Euclidean dendrograms put almost every class in one cluster, so the index is forced to zero. DINO-B does not fall in the island. Figure 5a shows the naive map.

Controlling the cut leaves a moderate gap. The criterion of Section 3.5 selects cosine distances with average linkage on ImageNet. Under it the island shrinks to a moderate gap, and about a third of the within-block agreement is missing under most admissible configurations. In cophenetic correlation the gap remains, and in triplet agreement, which needs no cut, it closes under cosine-average only. Figure 5b shows the selected map and Table 10 gives the six configurations.

The self-supervised tree is angular. DINOv2’s semantics concentrate in the angular component of its representation: for DINOv2-L and DINOv2-G, sibling triplets are resolved far better under cosine than under Euclidean distance. This angular tree is why the Euclidean dendrograms chain. The census repeated on cosine geometry agrees with the Euclidean census in almost every cell, and DINOv2 on ImageNet is genuine under both. Table 4 gives the cosine census.

Alignment with WordNet follows supervision and recipe. The correlation between inter-centroid and WordNet distances is highest for the contrastive VLMs, then the supervised ViTs, and lowest for DINOv2, with shuffle and Gaussian controls at zero. Supervision deepens the resemblance on average, and DeiT-B shows that the effect is recipe-dependent. The most aligned families carry the most fragile excess and DINOv2 the most robust one. Table 11 gives the correlations.

The recovery is not WordNet circularity. On DBpedia Classes, an ontology independent of WordNet, the three text embedders read a genuine excess and recover the true partition under every configuration. On HierarCaps, a four-level caption hierarchy with no class set, the encoders resolve sibling triplets far above chance and order the levels by projection radius. Table 11 gives both.

Text depends on recipe, scale and probe. On the ImageNet class names, 7 of 15 text models are genuine under the record. GPT-2 S is genuine at the margin, GPT-2 M is not, and GPT-2 L and XL are genuine under every reading, as are Pythia at every scale and OLMo-1B. The sentence embedders are absent on the class names and present on DBpedia, so the probe decides their verdict. Prompt template and extraction move the fragile values but not the scale ordering. Table 6 gives the census.

5.5 WHAT IS SHARED IS LOCAL

Models share neighborhoods, not metrics. Permutation-calibrated mutual-kNN agreement and centroid CKA survive across the sixty-six model pairs, every pair significant. The Poincaré readout improves neither, so models share who is near whom and differ in the metric, the reading of Gröger et al. (2026) on geometry. Table 12 gives the values.

6 IMPLICATIONS FOR HYPERBOLIC REPRESENTATION LEARNING

The raw reading cannot select a curvature. Khrulkov et al. (2020) set the curvature of the Poincaré ball from the relative hyperbolicity,

$$c = (0.144/\delta_{\text{rel}})^2, \quad \delta_{\text{rel}} = 2\delta/\text{diam}. \quad (7)$$

On the Gaussian band the rule assigns structureless clouds a curvature from 0.48 to 2.5, tracking the width of the backbone rather than its hierarchy. Table 3 gives the band.

The calibrated reading predicts the zero-cost gain. Within each hierarchical dataset the record excess predicts the nearest-centroid gain of the Poincaré readout on all four datasets and the few-shot gain on three. The depth verdict predicts no gain. The prediction describes the families it is scored on rather than validating a policy. Table 13 gives the correlations.

A change of readout collects what is there. Cosine collects the self-supervised structure, and the Poincaré readout adds +0.9 to +1.3 pp over cosine for the contrastive VLMs on prototype tasks and nothing consistent elsewhere. Nothing helps on flat labels, and sample-level tasks never benefit. Before imposing curvature, read the excess and its rank, the depth test, and the cosine reading. Table 13 gives the gains.

7 CONCLUSION AND LIMITATIONS

Conclusion. Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry. The premise of latent hyperbolicity is confounded, and its sample-level reading sits within null noise. Curvature should be measured rather than assumed, and a change of readout collects what is there.

Limitations. (i) The excess certifies structure beyond second moments, star-like clustering included, so depth needs the separate test. (ii) The depth test is conservative and weak at ImageNet’s noise level, certifies hierarchy above its frame only, leaves hierarchy within the clusters untested, and its one trained positive control is inconclusive. (iii) A causal LM’s class representation depends on the extraction protocol as much as on the model, and its values are template-conditioned. (iv) The census is class-centroid geometry on six datasets, and the sample-level reading covers two datasets and one image budget. (v) The objective–geometry link is correlational, resting chiefly on DINOv2’s scale range and on two leaf-label ViTs. (vi) Downstream scope is prototype-based metric tasks with modest gains.

REPRODUCIBILITY STATEMENT

Every number in the paper is annotated in the source with the result file that produces it, and all tables are generated by script from those files (Table 14). The full pipeline with fixed seeds will be released in the anonymized repository (TODO(author)). Its `tool/` directory ships the instrument as one script that reproduces any cell of Table 1 from its centroid matrix. Estimator settings are stated in Section 4 and the model panel in Table 5.

ETHICS STATEMENT

This work analyzes publicly released models and standard public datasets, involves no human subjects and no personal data, and proposes an analysis methodology and a metric-selection heuristic. We do not foresee ethical risks beyond those already associated with the underlying models and datasets.

AI USE STATEMENT

In this work, we used generative AI tools for writing assistance, for the retrieval of references, and for the implementation and execution of experiments under the authors’ direction, with LLM-simulated reviews used as methodological feedback. We have not used generative AI tools for other tasks with required disclosure. We have reviewed all AI-assisted work, and we take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

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(a) Quadruple budget: the excess at 10^4 to 2×10^6 sampled quadruples per seed under the record (Haar null, p99.9, 200 replicates).

model	data	protocol	10^4	5×10^4	10^5	5×10^5	10^6	2×10^6	drift ($\geq 10^5$)	drift/s.d.
ViT-L	IN	record	-0.0161	-0.0161	-0.0160	-0.0157	-0.0159	-0.0159	0.0002	0.07
DINOv2-L	IN	record	-0.0131	-0.0136	-0.0137	-0.0135	-0.0136	-0.0136	0.0002	0.38
CLIP-B	IN	record	-0.0063	-0.0060	-0.0059	-0.0065	-0.0064	-0.0064	0.0007	0.13
ViT-L	C100	record	-0.0208	-0.0208	-0.0206	-0.0205	-0.0205	-0.0205	0.0002	0.03
DINOv2-L	C100	record	-0.0348	-0.0348	-0.0348	-0.0348	-0.0347	-0.0347	0.0001	0.07
CLIP-B	C100	record	-0.0186	-0.0195	-0.0188	-0.0195	-0.0194	-0.0194	0.0006	0.10
ViT-L	DTD	record	-0.0305	-0.0308	-0.0312	-0.0311	-0.0313	-0.0313	0.0002	0.05
DINOv2-L	DTD	record	-0.0418	-0.0419	-0.0421	-0.0420	-0.0422	-0.0420	0.0002	0.06
CLIP-B	DTD	record	-0.0243	-0.0235	-0.0242	-0.0240	-0.0243	-0.0242	0.0003	0.05
ViT-L	C100	centroid bootstrap: excess	-0.0493, s.d. 0.0029	(30 resamples of the per-class images, 500 images/class)						
DINOv2-L	C100	centroid bootstrap: excess	-0.0493, s.d. 0.0022	(30 resamples of the per-class images, 500 images/class)						
DINOv2-G	C100	centroid bootstrap: excess	-0.0778, s.d. 0.0020	(30 resamples of the per-class images, 500 images/class)						
CLIP-B	C100	centroid bootstrap: excess	-0.0476, s.d. 0.0017	(30 resamples of the per-class images, 500 images/class)						
ViT-L	DTD	centroid bootstrap: excess	-0.0803, s.d. 0.0049	(30 resamples of the per-class images, 80 images/class)						
DINOv2-L	DTD	centroid bootstrap: excess	-0.0878, s.d. 0.0048	(30 resamples of the per-class images, 80 images/class)						
DINOv2-G	DTD	centroid bootstrap: excess	-0.0863, s.d. 0.0051	(30 resamples of the per-class images, 80 images/class)						
CLIP-B	DTD	centroid bootstrap: excess	-0.0581, s.d. 0.0054	(30 resamples of the per-class images, 80 images/class)						

Table 2: **The excess is budget-stable, the genuine count survives resampling and estimator noise, the reading follows the hierarchy of the labels rather than their number, and training moves it.** (a) Excess against the quadruple budget for nine cells; drift = range of the record excess over budgets $\geq 10^5$, divided by the excess’s spread in the last column; under the record every sign holds at every budget. (b) Bootstrap of the ImageNet excess over resampled images per class: the bootstrap s.d. is at most 0.0011 and every resample of every backbone is sign-negative.

A PROOFS

Proof of Proposition 1. (a) Each pairing sum in Eq. 1 adds two distances of the space, each of which lies in $[d_{\min}, d_{\max}]$. Hence $S_{\max} \leq 2d_{\max}$ and $S_{\min} \geq 2d_{\min}$, so the defect $\frac{1}{2}(S_{\max} - S_{\min})$ is at most $d_{\max} - d_{\min}$ for every quadruple, and so is its supremum δ . Dividing by the diameter $d_{\max}$ gives $\delta_{\text{norm}} \leq 1 - d_{\min}/d_{\max}$; the same bound holds for the 99.9th percentile of the sampled defects, which never exceeds their supremum. (b) Let $x_1, \dots, x_n$ be iid standard Gaussian in $\mathbb{R}^d$ with n fixed. For any pair $i \neq j$ the coordinates of $x_i - x_j$ are iid with variance 2, so $\|x_i - x_j\|^2/(2d) \rightarrow 1$ almost surely as $d \rightarrow \infty$ by the law of large numbers, and the convergence holds jointly over the finitely many pairs. Therefore $d_{\min}/d_{\max} \rightarrow 1$, and by part (a) $\delta_{\text{norm}} \rightarrow 0$. The rate is the distance-concentration rate of Beyer et al. (1999) and Aggarwal et al. (2001): the relative contrast $(d_{\max} - d_{\min})/d_{\min}$ vanishes as $d^{-1/2}$. For a cloud with covariance Σ the same argument runs with d replaced by the effective dimension $(\text{tr } \Sigma)^2/\text{tr}(\Sigma^2)$, and a maximum over a growing sample of quadruples is non-decreasing in the sample size. $\square$

B ONE TABLE PER QUESTION

Every table is generated by script from the released per-cell result files, and no value is hand-transcribed; the tables are numbered in the order the main text first cites them, one per question, with their result files indexed in Table 14.

B.1 IS THE READING ROBUST? QUADRUPLE BUDGET, NULL CONSTRUCTION, RESAMPLING AND CLASS COUNT

B.2 IS A LOW RAW READING EVIDENCE? CALIBRATION ON REFERENCE GEOMETRIES AND SYNTHETIC CLUSTERS

B.3 IS THE CLUSTERED STRUCTURE GENUINE? THE VISION CENSUS UNDER EVERY CONSTRUCTION

What each null preserves. The spectrum-matched null, Gaussian or Haar, preserves n , d , the mean and the covariance spectrum and destroys all higher-order structure; the PC-permutation null preserves each principal component’s marginal and destroys the dependence across components; the label shuffle keeps the point cloud itself and destroys the embedding–label correspondence; and the hub-randomizing null keeps each cluster’s geometry and hub assignment and destroys the hub arrangement, which is why it serves the depth test only when read against a matched star. “Genuine”

(b) ImageNet centroid bootstrap under the record: 30 resamples of the 100 training images per class (20 null replicates per resample).

model	excess (reference)	bootstrap mean	bootstrap s.d.	fraction negative
ViT-T	-0.0055	-0.0051	0.0011	1.00
ViT-S	-0.0126	-0.0122	0.0007	1.00
ViT-B	-0.0165	-0.0160	0.0006	1.00
ViT-L	-0.0160	-0.0160	0.0004	1.00
DINO-B	-0.0209	-0.0208	0.0003	1.00
DINOv2-S	-0.0164	-0.0162	0.0004	1.00
DINOv2-B	-0.0167	-0.0164	0.0002	1.00
DINOv2-L	-0.0136	-0.0134	0.0001	1.00
DINOv2-G	-0.0080	-0.0083	0.0004	1.00
CLIP-B	-0.0068	-0.0069	0.0009	1.00
CLIP-L	-0.0094	-0.0095	0.0006	1.00
SigLIP-B	-0.0080	-0.0081	0.0006	1.00

Table 2: (continued)

(c) Joint sensitivity of the genuine count: z_{joint} against null, bootstrap and estimator noise, and (in parentheses) the number of the 30 centroid resamples in which the cell is genuine.

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	-1.1 (0) ^o	-0.7 (0) ^o	-1.1 (0) ^o	-2.0 (13) [†]	+0.9 (0) ^o	-1.0 (0) ^o
ViT-S	-3.6 (30)	-0.0 (0) ^o	-2.7 (30)	-3.1 (30)	-1.7 (30) [†]	-1.3 (0) ^o
ViT-B	-5.2 (30)	+0.8 (0) ^o	-2.4 (30)	-4.0 (30)	-2.1 (30)	-0.6 (0) ^o
ViT-L	-5.0 (30)	-4.0 (30)	-2.8 (30)	-6.4 (30)	-2.5 (30)	-1.6 (0) ^o
DINO-B	-8.9 (30)	-7.6 (30)	-2.7 (30)	-4.5 (30)	-0.3 (0) ^o	-1.5 (0) ^o
DINOv2-S	-15.9 (30)	-9.7 (30)	-2.8 (30)	-3.2 (30)	-1.6 (30) ^{o‡}	-0.2 (0) ^o
DINOv2-B	-31.6 (30)	-14.7 (30)	-4.3 (30)	-8.1 (30)	-2.7 (30)	-0.6 (0) ^o
DINOv2-L	-30.7 (30)	-21.6 (30)	-4.9 (30)	-10.6 (30)	-1.8 (29) [†]	-0.5 (0) ^o
DINOv2-G	-11.5 (30)	-28.5 (30)	-4.8 (30)	-8.2 (30)	-2.0 (30)	-0.8 (0) ^o
CLIP-B	-1.3 (0) ^o	-3.1 (30)	-1.8 (7) [†]	-3.5 (30)	-1.3 (0) ^o	-0.0 (0) ^o
CLIP-L	-2.1 (12) [†]	-3.5 (30)	-2.0 (30) [†]	-4.3 (30)	-1.5 (0) ^o	-0.4 (0) ^o
SigLIP-B	-1.7 (0) ^o	-4.3 (30)	-2.6 (30)	-4.1 (30)	-2.0 (30) [†]	-1.9 (30) [†]

Table 2: (continued) (c) Per cell: $z_{\text{joint}} = \text{excess} / \sqrt{\sigma_{\text{null}}^2 + \sigma_{\text{boot}}^2 + \sigma_{\text{est}}^2}$ and the number of resamples in which the cell is genuine under BH over the 72 cells; ^o: not genuine in the record; [†]: record-genuine but failing one of the two criteria ($z_{\text{joint}} \leq -2$, genuine in ≥ 27 of 30); [‡]: not record-genuine but passing one. Counts: record 49/72 (18/24); $z_{\text{joint}} \leq -2$: 42/72 (18/24); genuine in ≥ 27 of 30 resamples: 47/72 (17/24). (d) Class-count control under the record (means over subset seeds); “below” = subset seeds whose real value lies below the null at uncorrected $p \leq 0.05$; NC adv = prototype-classifier advantage of the Poincaré readout in pp. (e) Raw δ within architecture: fine-tuning on a task without class hierarchy raises it and it falls across transformer depth.

(d) Class count against hierarchy depth: ImageNet subsets of $C = 100$ classes, random (spanning the hierarchy) or WordNet-coherent (siblings, effectively flat), means over five subset seeds; the other values of C are in the sweep file.

model	C	random subsets (span the hierarchy)				WordNet-coherent (siblings)			
		$\hat{\delta}_{99.9}$	excess	below	NC adv	$\hat{\delta}_{99.9}$	excess	below	NC adv
DINOv2-L	100	0.031	-0.032	5/5	+1.46	0.044	-0.031	5/5	+0.77

Table 2: (continued)

(e) Training moves the raw reading within an architecture: one row per intervention.

intervention	model	δ before	δ after	change
non-hierarchical fine-tuning	DINOv2-S	0.095	0.161	+69%
across depth (layer 1 $\rightarrow$ 12)	DINOv2-B	0.147	0.058	-61%

Table 2: (continued)

(a) Reference geometries.

space	absolute δ	δ_{norm}
balanced binary tree (depth 10)	0.000	0.000
$\mathbb{H}^2$ ($K = -1$), region radius $R=2/4/8/16$	0.65/0.69/0.69/0.69	0.162/0.087/0.043/0.022
uniform S^{99} (chord / geodesic)	0.25/0.37	0.143/0.179
iid Gaussian ($n=1000$), $d=192, \dots, 1536$	—	0.104/0.081/0.074/0.061/0.055/0.046

Table 3: **A low raw δ is not evidence of hierarchy, and the spectrum excess certifies clustering, not depth.** (a) Calibration on reference geometries: the estimator recovers the tree zero and the four-point constant of $\mathbb{H}^2$, spheres stay high, and iid Gaussians drift low as d grows, the dimension confound; the ordering trees $<$ hyperbolic $<$ spherical holds for hyperbolic regions of radius $R \geq 4$. Gaussian row from the released control file; tree/ $\mathbb{H}^2$ /sphere rows re-verified synthetically at every freeze. (b) Excess of synthetic clouds (means over seeds; within/between noise ratio in parentheses) under the spectrum null A, with Gaussian coefficients and with Haar-rotated coefficients (exact sample spectrum), and under the hub-randomizing null B, which keeps every cluster and resamples the hub configuration. A pure star sits far below null A under either construction, so null A certifies clustering, not depth; null B also reports the star as hierarchical, since a near-regular simplex of hubs already minimizes δ , which is why the depth test of Table 8 is read against a matched star rather than against null B. Under the Haar construction the two-level hierarchy sits $2\text{--}3\times$ below the star; under the plain Gaussian null the mid-radius star scores -0.104 and the 6×5 hierarchy -0.14 , a fraction of either’s excess.

(b) Synthetic clusters: what each null certifies.

synthetic cloud ($n=1000$, $d=768$)	$\hat{\delta}$	spectrum null A		hub-randomizing null B	
		Gaussian	Haar	Gaussian	Haar
30-cluster star, tight (0.1)	0.055	−0.115	−0.115	−0.126	−0.058
30-cluster star, mid (0.3)	0.059	−0.104	−0.103	−0.114	−0.049
30-cluster star, loose (0.6)	0.061	−0.084	−0.082	−0.090	−0.033
2-level hierarchy 6×5 (0.3)	0.044	−0.143	−0.143	−0.113	−0.112

Table 3: (continued)

(a) The census of record per cell: Haar spectrum-matched null, 99.9th-percentile statistic, 200 replicates.

model	IN			C100			C10			DTD			FMNIST			MNIST		
	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p
ViT-T	−0.005°	169	.159	−0.006°	151	.249	−0.021°	172	.144	−0.014	197	.020	+0.018°	34	.831	−0.026°	168	.164
ViT-S	−0.012	200	.005	−0.000°	111	.448	−0.056	200	.005	−0.019	199	.010	−0.033	196	.025	−0.037°	181	.100
ViT-B	−0.016	200	.005	+0.006°	43	.786	−0.050	200	.005	−0.021	200	.005	−0.042	200	.005	−0.013°	142	.294
ViT-L	−0.016	200	.005	−0.020	200	.005	−0.060	200	.005	−0.031	200	.005	−0.051	200	.005	−0.040°	193	.040
DINO-B	−0.020	200	.005	−0.032	200	.005	−0.061	200	.005	−0.027	200	.005	−0.007°	127	.368	−0.034°	188	.065
DINOv2-S	−0.016	200	.005	−0.031	200	.005	−0.071	200	.005	−0.018	200	.005	−0.031°	193	.040	−0.004°	112	.443
DINOv2-B	−0.017	200	.005	−0.031	200	.005	−0.112	200	.005	−0.035	200	.005	−0.062	200	.005	−0.017°	147	.269
DINOv2-L	−0.014	200	.005	−0.035	200	.005	−0.128	200	.005	−0.042	200	.005	−0.035	197	.020	−0.014°	132	.343
DINOv2-G	−0.008	200	.005	−0.040	200	.005	−0.130	200	.005	−0.039	200	.005	−0.046	199	.010	−0.022°	158	.214
CLIP-B	−0.007°	172	.144	−0.019	200	.005	−0.036	195	.030	−0.024	200	.005	−0.035°	181	.100	−0.001°	104	.483
CLIP-L	−0.009	197	.020	−0.018	200	.005	−0.043	199	.010	−0.027	200	.005	−0.037°	188	.065	−0.008°	126	.373
SigLIP-B	−0.008°	189	.060	−0.022	200	.005	−0.050	199	.010	−0.023	200	.005	−0.042	198	.015	−0.044	198	.015

Table 4: **Clustered structure is the rule at the class level under every construction of the null and statistic, and the verdict does not depend on the metric.** (a) Per cell: excess of $\hat{\delta}_{99.9}$ over the null mean, r = replicates above the real value, $p = (1 + \#\{\text{null} \leq \text{real}\})/201$; genuine = Benjamini–Hochberg-corrected $p \leq 0.05$ over the 72 cells (°: not genuine). Genuine cells: 49/72 overall, 18/24 on ImageNet+CIFAR-100; sign agreement with the original 5-replicate Gaussian $\times$ supremum census on the 72 comparable cells: 67/72.

in the text always means beyond the first of these. In plain terms, the spectrum null is a cloud with the same shape as the real one and no structure inside it; the hub null of Section 5 keeps every cluster of the real cloud and only rearranges where the clusters sit.

(c) The same census on cosine geometry (L2-normalized centroids, geodesic distances, Haar null on the normalized cloud): robustness, not the record.

model	IN		C100		C10		DTD		FMNIST		MNIST	
	exc.	r	exc.	r	exc.	r	exc.	r	exc.	r	exc.	r
ViT-T	-0.012	200	-0.014 ^o	191	-0.043	198	-0.017	199	+0.007 ^o	73	-0.026 ^o	167
ViT-S	-0.017	200	-0.017	200	-0.073	200	-0.021	200	-0.049	200	-0.036 ^o	183
ViT-B	-0.022	200	-0.006 ^o	170	-0.066	200	-0.021	200	-0.041	199	-0.010 ^o	130
ViT-L	-0.021	200	-0.024	200	-0.080	200	-0.024	200	-0.066	200	-0.043	196
DINO-B	-0.021	200	-0.035	200	-0.070	200	-0.024	200	-0.006 ^o	116	-0.038 ^o	193
DINOv2-S	-0.014	200	-0.033	200	-0.084	200	-0.018	200	-0.054	200	-0.004 ^o	113
DINOv2-B	-0.013	200	-0.032	200	-0.140	200	-0.021	200	-0.081	200	-0.019 ^o	147
DINOv2-L	-0.010	200	-0.029	200	-0.151	200	-0.024	200	-0.051	200	-0.020 ^o	145
DINOv2-G	-0.006	200	-0.029	200	-0.169	200	-0.017	200	-0.064	200	-0.023 ^o	157
CLIP-B	-0.017	200	-0.023	200	-0.039	197	-0.031	200	-0.038 ^o	185	+0.002 ^o	93
CLIP-L	-0.017	200	-0.025	200	-0.045	199	-0.031	200	-0.038 ^o	192	-0.008 ^o	124
SigLIP-B	-0.013	200	-0.025	200	-0.053	200	-0.027	200	-0.052	199	-0.044	197

Table 4: (continued) (c) Cosine reading, BH over 72 cells: sign-negative 70/72, genuine 56/72 and 22/24 on ImageNet+CIFAR-100, verdict agreement with the Euclidean record 65/72 cells. Text under the same cosine protocol (padding-free embeddings): genuine 8/15: gpt2-m, gpt2-l, gpt2-xl, pythia-410m, pythia-1b, pythia-2b8, olmo-1b, gte-qwen2. The angular tree: sibling-triplet agreement on CIFAR-100 for DINOv2-L is 0.91 under cosine against 0.71 under Euclidean distance. (d) Two self-supervised ResNet-50 backbones read the same clustered structure as the ViTs.

(d) Backbones outside the ViT panel on the ImageNet centroid store (spectrum null, 20 replicates).

model	d	$\hat{\delta}$	excess (z)	reps above
Barlow Twins (ResNet-50)	2048	0.075	-0.039 (-4.8)	20/20
BYOL (ResNet-50)	2048	0.073	-0.020 (-2.4)	20/20

Table 4: (continued)

B.4 THE MODEL PANEL AND EXTRACTION

B.5 IS TEXT THE SAME? THE TEXT CENSUS UNDER THE RECORD

Extraction, anisotropy and templates. Extraction is part of the measurement: left-padded batching moves GPT-2’s $\hat{\delta}$ by up to 0.018, of the order of the fragile excesses; the padding-free protocol is the record. Genuineness is not a proxy for anisotropy: GTE-base has the highest effective rank and is not genuine, and the only LLM-backed embedder tested (GTE-Qwen2-1.5B) is not genuine under the record. Causal-LM values are template-sensitive but the scale ordering is template-robust (Table 6).

B.6 DOES THE PREMISE SURVIVE AT THE SAMPLE LEVEL? THE CENSUS ON PER-IMAGE FEATURES

B.7 IS THE STRUCTURE DEEPER THAN A STAR? THE DEPTH TEST

Real-data readings by regime. ImageNet ($n=1000$, WordNet $K=30$) lies in the validated regime: the real cloud is more hierarchical than its anisotropic matched star beyond the noise ($z \leq -2$) in 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L), never less, under Gaussian and under Haar-resampled star hubs alike (Table 8, Figure 4a). CIFAR-100 ($n=100$, $K=20$) lies in the unvalidated regime, where pure stars are declared more hierarchical in 17% of runs; its readings (3 of 12 with $z \leq -2$, none with $z \geq +2$) are reported without certification (Table 8). Validity domain of the test as run: $n \approx 1000$; at $n=100$ the false-alarm rate is 0% with $K=6$ (17 points per cluster) and 17% with $K \geq 12$, so the test is not validated below about ten points per cluster. The bar was fixed before the sweep; the restriction to the validated regime was adopted after it.

A trained control, inconclusive. We fine-tuned the census ViT-B/16 for two passes over the images behind its ImageNet centroids with two objectives on the same batches: cross-entropy alone, and cross-entropy plus a hierarchical cross-entropy over the WordNet superclasses. Both fine-tuned models keep the census excess of the frozen backbone (-0.015 and -0.016 against -0.016) and both are certified by the depth test, with z of -2.33 and -2.77 against -3.62 for the frozen back-

type	model	params
Supervised ViT (vision)	ViT-T (i21k)	5M
Supervised ViT (vision)	ViT-S (i21k)	22M
Supervised ViT (vision)	ViT-B (i21k)	86M
Supervised ViT (vision)	ViT-L (i21k)	307M
SSL (vision)	DINO-B	86M
SSL (vision)	DINOv2-S	22M
SSL (vision)	DINOv2-B	86M
SSL (vision)	DINOv2-L	307M
SSL (vision)	DINOv2-G	1.1B
Contrastive (vision)	CLIP-B	86M
Contrastive (vision)	CLIP-L	307M
Contrastive (vision)	SigLIP-B	86M
Causal LM	GPT-2 S	117M
Causal LM	GPT-2 M	345M
Causal LM	GPT-2 L	774M
Causal LM	GPT-2 XL	1.5B
Causal LM	Pythia-410M	410M
Causal LM	Pythia-1B	1.0B
Causal LM	Pythia-2.8B	2.8B
Causal LM	OLMo-1B	1.0B
Causal LM	OLMo-7B	7.0B
Text embedder	BGE-base	110M
Text embedder	BGE-large	335M
Text embedder	GTE-base	110M
Text embedder	GTE-large	335M
Text embedder	GTE-Qwen2-1.5B	1.5B
Text embedder	E5-base	110M
Text embedder	E5-large	335M
Controls (vision)	DeiT-B/16 (IN-1k, no distillation)	86M
Controls (vision)	ViT-B/16 augreg (IN-1k)	86M
Controls (vision)	MERU ViT-S/B/L and CLIP twins	22M/86M/307M
Controls (vision)	Barlow Twins, BYOL (ResNet-50)	24M

Table 5: The model panel. 12 vision backbones (all in the census; the 10 with cached per-dataset features, all but DINO-B and SigLIP-B, form the task grid), 9 causal LMs, 7 text embedders, and the vision models used only as controls. Class centroids average 100 cached training images per class on ImageNet and the full cached split, 80–6,000 images per class, on the other datasets. Feature extraction: vision backbones use the encoder’s default pooled embedding (timm, num_classes=0, CLS or global pool per architecture); causal LMs use the last-token hidden state, extracted one prompt at a time (padding-free: left-padded batching alters GPT-2’s hidden states, Table 6); text embedders use their recommended pooling (CLS for BGE, mean for GTE/E5, last token for GTE-Qwen2); CLIP/SigLIP text towers use their text encoders.

bone: the hierarchical term lowers z relative to plain cross-entropy but neither reaches the frozen reading, so the control neither confirms nor refutes the test (Table 8). The full training set could not be read at training speed from the archive, so the schedule is two passes over the census subset.

B.8 HOW MUCH DEPTH CAN THE TEST SEE? POWER ON SYNTHETIC AND IMPLANTED HIERARCHIES

B.9 IS THE ISLAND REAL? THE TREE MAP UNDER SIX CONFIGURATIONS

Threshold sensitivity of the selection criterion of §5.4: the selected configuration is stable for degeneracy bounds in $[0.45, 0.50]$ on ImageNet and $[0.40, 0.50]$ on CIFAR-100; tighter bounds select cosine-complete on ImageNet (gap 0.13) and cosine-Ward on CIFAR-100 (gap 0.56), and in every case the label-free gap is moderate and configuration-sensitive, never a categorical island. The criterion must be applied per dataset: the configuration it selects on DBpedia (Euclidean, average linkage, where nothing is degenerate) is exactly the degenerate naive configuration on ImageNet, so a criterion transferred across datasets would reintroduce the island.

B.10 WHOSE TAXONOMY? WORDNET ALIGNMENT, SUPERCLASS RECOVERY, DBPEDIA AND HIERARCAPS

On HierarCaps (Alper & Averbuch-Elor, 2024) (1000 four-level caption chains), text encoders show the same reading with no class set: the projection radius orders levels within a chain, sibling triplets are resolved at 0.83–0.97 (chance 0.5), and the leaf excess repeats the text census (Table 11).

(a) The census of record on the 1000 ImageNet class prompts: padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates.

model	type	d	$\hat{\delta}_{99.9}$	excess	record		p	2×2	cosine	
					exc./null	$r/200$			exc.	r
GPT-2 S	causal LM	768	0.077	-0.013	-15%	197	.020	●○○○	+0.000°	101
GPT-2 M	causal LM	1024	0.052	-0.005°	-9%	170	.154	○○○○	-0.019	200
GPT-2 L	causal LM	1280	0.053	-0.023	-30%	200	.005	●●●●	-0.016	200
GPT-2 XL	causal LM	1600	0.048	-0.027	-36%	200	.005	●●●●	-0.018	200
Pythia-410M	causal LM	1024	0.067	-0.018	-21%	200	.005	●●●●	-0.024	200
Pythia-1B	causal LM	2048	0.066	-0.015	-18%	199	.010	●●●●	-0.026	200
Pythia-2.8B	causal LM	2560	0.066	-0.017	-20%	200	.005	●●●●	-0.023	200
OLMo-1B	causal LM	2048	0.051	-0.024	-32%	200	.005	●●●●	-0.032	200
OLMo-7B	causal LM	4096	not re-extracted (exceeds the local GPU)							
BGE-base	embedder	768	0.078	+0.003°	+4%	26	.871	○○○○	-0.002°	151
BGE-large	embedder	1024	0.078	+0.002°	+3%	49	.756	○○○○	-0.003°	170
GTE-base	embedder	768	0.076	+0.004°	+6%	11	.945	○○○○	+0.001°	79
GTE-large	embedder	1024	0.075	+0.003°	+3%	36	.821	○○○○	-0.001°	131
GTE-Qwen2-1.5B	embedder	1536	0.078	+0.002°	+3%	58	.711	○○○○	-0.006	199
E5-base	embedder	768	0.074	+0.004°	+6%	15	.925	○○○○	+0.000°	97
E5-large	embedder	1024	0.071	+0.002°	+3%	44	.781	○○○○	-0.001°	119

Table 6: **In text, clustered structure depends on recipe and scale: genuine at every Pythia scale and at the larger GPT-2 sizes, at the margin for GPT-2 S, absent for GPT-2 M and for the sentence embedders on this probe.** (a) r = replicates above the real value, p the left-tail add-one p -value; genuine = BH-corrected $p \leq 0.05$ over the 15 models (°: not genuine); exc./null: excess as a fraction of the null reading; the 2×2 column gives the verdicts under Haar $\times$ p99.9, Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum (● genuine); cosine: the same census on cosine geometry. Genuine 7/15; GPT-2 S is genuine at $p = .020$ with excess -0.013 and GPT-2 M is not ($p = .154$, -0.005), while L and XL are genuine under every construction. GTE-base has the highest effective rank and is not genuine, so genuineness is not a proxy for anisotropy; the only LLM-backed embedder (GTE-Qwen2-1.5B) is not genuine. (b) Causal-LM values are template-sensitive (range of raw $\hat{\delta}$ up to 0.065 for causal LMs against ≤ 0.012 for embedders; “gloss only” contains no class name) but the scale ordering is template-robust: with only 3 null replicates per cell the starred S and M cells are borderline, as the 200-replicate record in (a) shows, whereas the L/XL cells are not in doubt. (c) Precision is irrelevant (fp16 equals fp32 to four decimals); for GPT-2, left-padded batching changes the hidden states (mean cosine to the padding-free extraction 0.98) and moves $\hat{\delta}$ by up to 0.018, while OLMo handles left padding correctly and is invariant, which is why padding-free extraction is the record.

(b) Prompt templates: raw $\hat{\delta}$ per template and, for the four GPT-2 sizes, the template-conditioned excess over the spectrum null (3 replicates; *: $|z| \geq 2$) and its fraction of the null.

template	raw $\hat{\delta}$				GPT-2 excess (padding-free)				GPT-2 excess / null			
	BGE-b	E5-b	GPT-2 S	GPT-2 M	S	M	L	XL	S	M	L	XL
photo	0.124	0.114	0.131	0.091	-0.016*	-0.019*	-0.043*	-0.050*	-11%	-19%	-34%	-41%
name_only	0.118	0.119	0.145	0.122	-0.006	-0.007	-0.048*	-0.051*	-6%	-5%	-39%	-43%
image	0.125	0.117	0.110	0.110	-0.018*	-0.028*	-0.050*	-0.049*	-13%	-26%	-41%	-41%
closeup	0.126	0.120	0.086	0.100	+0.007	-0.018*	-0.044*	-0.049*	+5%	-18%	-36%	-40%
this_is	0.125	0.115	0.080	0.134	-0.032*	-0.022*	-0.042*	-0.046*	-19%	-22%	-34%	-38%
wild	0.115	0.116	0.089	0.104	-0.023*	-0.067*	-0.014	-0.010*	-16%	-43%	-11%	-8%
def_pair	0.119	0.108	0.105	0.081	-0.002	-0.019*	-0.036*	-0.035*	-1%	-21%	-28%	-29%
gloss_only	0.115	0.115	0.124	0.107	-0.020*	-0.030*	-0.033*	-0.036*	-14%	-31%	-27%	-29%
concept	0.119	0.120	0.112	0.112	-0.002	-0.003	-0.043*	-0.041*	-1%	-3%	-38%	-38%
discussion	0.121	0.116	0.105	0.098	-0.010	-0.013*	-0.041*	-0.040*	-7%	-17%	-35%	-35%
mean over templates					-0.012	-0.023	-0.040	-0.041				
templates at $ z \geq 2$					5 of 10	8 of 10	9 of 10	10 of 10				

Table 6: (continued)

(c) Extraction: the same 1000 prompts, only batch size and precision vary.

batch size	GPT-2 M		OLMo-1B	
	fp32 $\hat{\delta}$	fp16 $\hat{\delta}$	fp32 $\hat{\delta}$	fp16 $\hat{\delta}$
1 (no padding)	0.0816	0.0817	0.0817	0.0817
16	0.0996	0.0997	0.0817	0.0816
32	0.0905	0.0905	0.0817	0.0817

Table 6: (continued)

model	CIFAR-100 (10 images/class, $n=1000$)					DTD (22 images/class, $n=1034$)				
	sup. $\hat{\delta}$	$\hat{\delta}_{99.9}$	excess	exc./null	$r/200$ (p)	sup. $\hat{\delta}$	$\hat{\delta}_{99.9}$	excess	exc./null	$r/200$ (p)
ViT-T	0.130	0.093	-0.0016°	-2%	134 (0.333)	0.124	0.087	-0.0004°	-0%	111 (0.448)
ViT-S	0.121	0.086	+0.0051°	+6%	21 (0.896)	0.103	0.070	-0.0062	-8%	197 (0.020)
ViT-B	0.121	0.082	+0.0031°	+4%	50 (0.751)	0.092	0.060	-0.0080	-12%	200 (0.005)
ViT-L	0.100	0.067	-0.0041°	-6%	185 (0.080)	0.092	0.056	-0.0073	-12%	200 (0.005)
DINO-B	0.104	0.070	-0.0042°	-6%	194 (0.035)	0.110	0.075	+0.0008°	+1%	80 (0.602)
DINOv2-S	0.100	0.065	-0.0049	-7%	200 (0.005)	0.145	0.090	+0.0058°	+7%	0 (1.000)
DINOv2-B	0.089	0.050	-0.0090	-15%	200 (0.005)	0.139	0.073	-0.0019°	-3%	162 (0.194)
DINOv2-L	0.087	0.043	-0.0131	-24%	200 (0.005)	0.131	0.064	-0.0060	-9%	200 (0.005)
DINOv2-G	0.102	0.036	-0.0232	-39%	200 (0.005)	0.136	0.065	-0.0079	-11%	200 (0.005)
CLIP-B	0.117	0.080	-0.0060°	-7%	184 (0.085)	0.125	0.089	+0.0021°	+2%	71 (0.647)
CLIP-L	0.132	0.084	+0.0037°	+5%	35 (0.826)	0.120	0.082	-0.0008°	-1%	124 (0.383)
SigLIP-B	0.113	0.078	-0.0074	-9%	199 (0.010)	0.114	0.079	-0.0044°	-5%	184 (0.085)

Table 7: The sample-level reading, the object of prior latent-hyperbolicity claims, sits within null noise in most cells. The instrument on ≈ 1000 stratified training images per cell (per-image features, not centroids): sup. $\hat{\delta}$ is the raw supremum δ_{norm} the literature reports; $\hat{\delta}_{99.9}$, excess, exc./null, $r/200$ and p follow Table 1 (Haar spectrum-matched null on the sample cloud, 99.9th-percentile statistic, 200 replicates); °: not genuine under Benjamini–Hochberg over the 24 cells. The raw supremum lies in the band 0.087–0.145; the excess is not genuine in 14 of 24 cells and genuine in 10 (DINOv2-S, DINOv2-B, DINOv2-L, DINOv2-G, SigLIP-B on CIFAR-100; ViT-S, ViT-B, ViT-L, DINOv2-L, DINOv2-G on DTD), where it reaches at most -0.023 , that is -39% to -7% of the null reading.

B.11 IS LOCAL STRUCTURE SHARED? PERMUTATION-CALIBRATED AGREEMENT

B.12 WHAT FOLLOWS FOR PRACTICE? THE COROLLARY

One parameter-free projection and two tasks serve the corollary. The parameter-free map $\phi(x) = \tanh(\frac{1}{2}\|s(x-\mu)\|) \cdot \frac{s(x-\mu)}{\|s(x-\mu)\|}$ centers features, scales the 95th-percentile norm to radius $1/\sqrt{2}$, and preserves directions; prototypes in the ball are means of projected points re-mapped through the origin. Tasks: nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes); metric arms: Euclidean (R), Poincaré (H), the radial map with Euclidean distances (RT), cosine (COS), and the L2-normalized stack. Changing the readout metric on the intact configuration, which is what the rule selects among, is distinct from deforming the configuration itself.

The calibrated reading predicts the gain within datasets. Within each hierarchical dataset the Poincaré advantage tracks the record excess: the nearest-centroid correlation is -0.73 on ImageNet with a confidence interval that excludes zero on all four datasets, the few-shot correlation excludes zero on CIFAR-100, CIFAR-10 and DTD but not on ImageNet, and the depth-test z predicts no gain anywhere (Table 13). The raw reading correlates as strongly (-0.84 on ImageNet for nearest-centroid) and survives partialing out dimension, baseline accuracy, family and spectral covariates; without DINOv2 it holds on ImageNet and CIFAR-100 but not on CIFAR-10/DTD. Pooled cross-dataset correlations are family-driven and not used as evidence; on flat-label datasets no metric helps. Gains carry paired confidence intervals and a McNemar test.

The training objective picks the metric. The radial map with Euclidean distances changes nothing, so the gain is the metric itself, and the split follows the objective: cosine suffices for DINOv2, Poincaré $\approx$ cosine for supervised ViTs, and Poincaré beats cosine for CLIP by up to $+1.3\text{pp}$ even after L2 normalization (Table 13). The rule is read off the families it is scored on: it describes them rather than validating a policy.

(a) The twelve backbones: depth = excess B of the real centroids minus that of a matched star (negative = more hierarchical above the frame), with z against the combined spread.

model	C100, $K=20$, Gaussian hubs		IN, $K=30$, Gaussian hubs		IN, $K=30$, Haar-resampled hubs			bal. frame
	iso. star	aniso. star	iso. star	aniso. star	star B	depth (z)	r_{star}	real z
ViT-T	+0.000 (+0.0)	-0.025 (-1.4)	-0.015 (-1.5)	-0.018 (-1.5)	+0.0011	-0.0184 (-1.45)	0/10	-2.1
ViT-S	+0.036 (+2.3)	-0.008 (-0.5)	-0.021 (-1.9)	-0.028 (-2.4)	+0.0010	-0.0257 (-2.04)	0/10	-1.7
ViT-B	+0.063 (+4.5)	+0.003 (+0.2)	-0.026 (-2.9)	-0.033 (-3.6)	+0.0012	-0.0347 (-3.01)	0/10	-2.8
ViT-L	+0.040 (+4.2)	-0.017 (-1.5)	-0.024 (-3.2)	-0.033 (-4.0)	+0.0000	-0.0337 (-3.73)	0/10	-5.6
DINO-B	+0.008 (+0.9)	-0.038 (-4.2)	-0.002 (-0.2)	-0.009 (-0.9)	+0.0002	-0.0092 (-1.02)	0/10	-1.0
DINOv2-S	+0.006 (+0.6)	-0.020 (-1.6)	+0.011 (+0.9)	+0.008 (+0.7)	+0.0007	+0.0083 (+0.71)	10/10	+0.1
DINOv2-B	+0.041 (+5.4)	+0.010 (+0.9)	-0.011 (-0.9)	-0.016 (-1.2)	+0.0018	-0.0155 (-1.18)	0/10	-0.3
DINOv2-L	+0.042 (+6.8)	+0.013 (+1.2)	-0.024 (-2.3)	-0.028 (-2.4)	-0.0007	-0.0267 (-2.21)	0/10	-1.7
DINOv2-G	+0.026 (+3.8)	-0.007 (-0.7)	-0.024 (-1.6)	-0.029 (-1.8)	-0.0021	-0.0263 (-1.50)	0/10	-0.9
CLIP-B	+0.019 (+1.8)	-0.024 (-2.0)	-0.006 (-0.7)	-0.016 (-2.0)	+0.0012	-0.0171 (-1.48)	1/10	-1.3
CLIP-L	+0.036 (+5.7)	-0.013 (-2.3)	+0.004 (+0.5)	-0.008 (-0.8)	+0.0018	-0.0090 (-0.74)	2/10	-0.7
SigLIP-B	+0.027 (+2.8)	-0.021 (-2.2)	+0.001 (+0.0)	-0.011 (-0.9)	+0.0024	-0.0121 (-0.78)	1/10	-1.1

Table 8: **Hierarchy above the superclasses is certified in four ImageNet backbones under both matched stars, imposing the geometry does not create it, and leaf-label supervision can produce it or not.** (a) Isotropic star: Gaussian clouds with each superclass’s RMS spread (the earlier, withdrawn construction); anisotropic star: within each superclass a Haar sample with the cloud’s own covariance, hubs Gaussian at the real hub radius; Haar-resampled hubs: the same clusters with the hubs drawn as a Haar resample of the real hubs. Hub-randomizing Haar null, 10 star seeds; frames: CIFAR-100 coarse labels, ImageNet WordNet cut. The test is validated at $n=1000$ only (Table 9); CIFAR-100 readings ($n=100$) are reported without certification. Certified on ImageNet ($n=1000$, the validated regime): 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L) with z from -2.4 to -4.0 ; the withdrawn isotropic star reached $z=+6.8$ on CIFAR-100. With the hubs of the matched star drawn as a Haar resample of the real hubs (exact hub spectrum, 10 star seeds) the certified set is ViT-S, ViT-B, ViT-L, DINOv2-L, with z from -2.04 to -3.73 ; r_{star} counts the star seeds whose excess B lies at or below the real one, a rank whose resolution is one part in eleven and which does not separate the certified backbones from the others. Balanced frame (complete-linkage WordNet cut with the smallest variance of cluster sizes, chosen before any depth run): certified ViT-T, ViT-B, ViT-L. K sweep: CIFAR-100 $K=5$: iso 0/12 at $z \leq -2$, 7/12 at $z \geq +2$; aniso 0/12 at $z \leq -2$, 0/12 at $z \geq +2$. CIFAR-100 $K=10$: iso 0/12 at $z \leq -2$, 9/12 at $z \geq +2$; aniso 1/12 at $z \leq -2$, 0/12 at $z \geq +2$. CIFAR-100 $K=20$: iso 0/12 at $z \leq -2$, 8/12 at $z \geq +2$; aniso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=10$: iso 0/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 2/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=30$: iso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 4/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=60$: iso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 4/12 at $z \leq -2$, 0/12 at $z \geq +2$. (b) Supervision on leaf labels can produce the depth (the augreg recipe is certified under both stars) and need not (DeiT-B is not detected): the hierarchical label set of IN-21k is not what makes the census ViTs deep; their WordNet alignment is in Table 11.

(b) Two ViT-B/16 supervised on ImageNet-1k leaf labels only, through the census protocol, against the IN-21k census backbone.

model	IN excess (r)	C100 excess (r)	depth, Gaussian hubs (z)	depth, Haar hubs (z)
DeiT-B, IN-1k	-0.0073 (200)	-0.0257 (200)	-0.0213 (-1.26)	-0.0197 (-1.21)
ViT-B, augreg, IN-1k	-0.0070 (199)	-0.0117 (187)	-0.0249 (-3.03)	-0.0290 (-2.65)
ViT-B, augreg, IN-21k (census)	-0.0162 (200)	+0.0057 (43)	-0.0335 (-3.62)	-0.0347 (-3.01)

Table 8: (continued)

A diagnostic, not a policy. Khrulkov et al. (2020) estimate $c \approx 0.33$ from the raw reading of their datasets; the same rule on structureless Gaussian clouds gives the curvatures of Section 6. Cosine is a safe default, gating on the reading is the weakest policy tested, and gains are modest and limited to prototype-based tasks (sample-level tasks never benefit, Table 7). On DBpedia, a three-level text hierarchy, the diagnostic predicts correctly on both sides: genuine excess, no metric gain.

B.13 PROVENANCE

(c) A hyperbolic backbone and its Euclidean twin under the census of record and the depth test.

		CIFAR-100 ($K=20, n=100$: depth unvalidated)			ImageNet ($K=30, n=1000$: depth validated)		
model	input	exc. (r)	$\hat{\delta}_{99.9}$ nat./Eucl.	depth z	exc. (r)	$\hat{\delta}_{99.9}$ nat./Eucl.	depth z
MERU ViT-S	images	−0.020 (199)	0.117/0.117	−1.5	−0.008° (186)	0.088/0.088	−1.5
CLIP ViT-S	images	−0.028 (200)	0.106/0.107	−3.9	−0.006° (171)	0.110/0.088	−1.8
MERU ViT-B	images	−0.020 (200)	0.114/0.114	−2.2	−0.010 (196)	0.087/0.087	−0.9
CLIP ViT-B	images	−0.030 (200)	0.105/0.103	−3.3	−0.009° (187)	0.107/0.085	−1.0
MERU ViT-L	images	−0.012° (195)	0.118/0.118	−0.3	−0.007° (184)	0.088/0.088	−0.5
CLIP ViT-L	images	−0.016 (199)	0.123/0.112	−1.7	−0.010 (197)	0.104/0.084	−1.2
MERU ViT-S	prompts	−0.022 (200)	0.086/0.086	−0.7	−0.003° (155)	0.085/0.086	+0.0
CLIP ViT-S	prompts	−0.023 (200)	0.085/0.082	−0.8	+0.006° (15)	0.093/0.087	+0.8
MERU ViT-B	prompts	−0.012 (200)	0.096/0.096	−0.7	−0.001° (132)	0.089/0.090	+0.8
CLIP ViT-B	prompts	−0.017 (200)	0.094/0.088	+0.3	+0.008° (2)	0.100/0.092	+0.2
MERU ViT-L	prompts	−0.006° (190)	0.102/0.102	+0.6	+0.001° (75)	0.093/0.093	+0.5
CLIP ViT-L	prompts	−0.012 (200)	0.098/0.092	+0.5	+0.006° (5)	0.101/0.092	+0.4

Table 8: (continued) (c) MERU (Desai et al., 2023) embeds images and text on the Lorentz hyperboloid with an entailment objective; its released twin is a CLIP baseline with the same ViT, RedCaps data and recipe minus the hyperbolic lift and entailment loss. Both pass through the instrument unchanged: class centroids of the projected embeddings (MERU: space-like hyperboloid coordinates; CLIP: unit vectors), Euclidean distances; exc. (r): excess over the Haar spectrum-matched null with the number of 200 replicates above the real value (°: not genuine under BH over the 24 cells); nat./Eucl.: the same statistic in the model’s own metric (Lorentz distance between tangent-space-mean centroids for MERU, angular for CLIP) against the Euclidean one; depth z : the anisotropic matched-star test of panel (a) (10 star seeds), validated at $n=1000$ only. On ImageNet images the excess runs from -0.007 to -0.010 for MERU and from -0.006 to -0.010 for its twin across ViT-S/B/L; MERU’s Lorentz-metric $\hat{\delta}_{99.9}$ differs from its Euclidean value by at most 0.0002; depth z runs from -1.5 to -0.5 for MERU and from -1.8 to -1.0 for CLIP, so neither is more hierarchical than a matched star and MERU is never ahead of its twin. Its embeddings sit in the near-flat regime of the hyperboloid: learned curvature $c=0.10$, spatial norm times $\sqrt{c}$ at the median 0.258–0.279 (95th percentile 0.263–0.287), and Lorentz-to-Euclidean pairwise distance ratio 0.9972–0.9991 at the median (0.9988–0.9997 on the class centroids), so the control tests the training objective, not a curved geometry. MERU’s hierarchy is a generic→specific (text⊃image) partial order, not a class taxonomy. (d) Two passes over the 100 training images per class behind the ImageNet centroids (patch embedding frozen, AdamW, same batches and seed for both objectives), with plain cross-entropy or with an added hierarchical cross-entropy over the WordNet 30-cut; both fine-tuned models stay certified and neither reaches the frozen reading, so the control is inconclusive.

(d) A trained control: the census ViT-B/16 fine-tuned with and without a hierarchical objective.

ViT-B/16	$\hat{\delta}_{99.9}$	excess	$r/200$	depth	z
frozen (census)	0.054	−0.0162	200	−0.0335	−3.62
cross-entropy	0.048	−0.0153	200	−0.0303	−2.33
cross-entropy + hierarchical CE (WordNet-30)	0.048	−0.0162	200	−0.0295	−2.77

Table 8: (continued)

(a) Synthetic hierarchies at $d=768$: two- and three-level hierarchies (pooled) and pure stars, within/between noise ratios 0.1/0.3/0.6, isotropic and anisotropic clusters pooled, 5 seeds.

frame	n	K	power @ ratio 0.1	0.3	0.6	star $z \leq -2$	star $z \geq +2$
top-level, isotropic star	100	6	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	100	12	0.00	0.00	0.00	0.00	0.13
top-level, isotropic star	100	20	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	100	30	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	6	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	12	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	20	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	30	0.00	0.00	0.00	0.00	0.17
leaf clusters, anisotropic star	100	6	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	100	12	1.00	1.00	1.00	0.17	0.00
leaf clusters, anisotropic star	100	20	1.00	1.00	1.00	0.17	0.00
leaf clusters, anisotropic star	1000	6	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	1000	12	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	1000	20	1.00	1.00	0.90	0.00	0.00
leaf clusters, anisotropic star	1000	30	1.00	1.00	0.70	0.00	0.00

Table 9: **The depth test meets its bar on synthetic hierarchies at ImageNet’s class count, raises no false alarm on real clouds under either star, and misses implanted depth at the real within-cluster spread.** (a) Power = fraction of hierarchy runs with $z \leq -2$; star columns = false-alarm rates in each direction. Top-level frame: the test receives the K super-cluster labels, as the earlier isotropic test did on the real backbones; its hub null keeps every frame cluster intact and only rearranges the K hubs, so hierarchy below the frame is invisible by construction. Leaf frame: the test receives the finest cluster labels with the anisotropic matched star (configurations with more leaves than points are infeasible and omitted). Pre-set bar: power ≥ 0.8 for two- and three-level hierarchies at noise ratios ≤ 0.3 for both n , and false alarms $\leq 5\%$ in each direction. Leaf frame: met at $n=1000$ (power 1.00 at ratios ≤ 0.3 and 0.90 at 0.6; false alarms 0% and 0%), failed at $n=100$, where $K \geq 12$ (under ten points per cluster) gives 17% false alarms at $z \leq -2$. Top-level frame: power 0.00, false alarms 16% pooled. (b) Six super-hubs of five, hub = s -super-hub + $(1 - s + 0.4s)$ -own Gaussian draw, hub cloud rescaled to the real RMS radius; the depth test of Table 8 runs unchanged, 5 implant seeds per s . hits: implant seeds with $z \leq -2$ (of 5); s^* : first s with ≥ 4 of 5; within/between: RMS of the within-cluster offsets over the point-weighted RMS of the hub displacements (1.3 to 3.9 on the real clouds, against 0.1–0.6 in the synthetic sweep); tight: the same implant after shrinking the offsets to within/between = 0.6 (2 seeds). Power (fraction of runs with $z \leq -2$) by s : $s=0$: 0.00, $s=0.25$: 0.00, $s=0.5$: 0.00, $s=0.75$: 0.00, $s=1$: 0.05; false alarms at $s=0$: 0 of 60. Under the Haar-hub star the same implants give 0 of 60 false alarms at $s=0$ and 1 of 60 detections at $s=1$ (power 0.02). Balanced frame: the $K=30$ agglomerative cut of the WordNet distance matrix (complete linkage) with the smallest variance of cluster sizes, chosen and logged before any depth run; sizes 1–154 classes against 1–316 for the frame of record.

(b) Implanted two-level trees on the real ImageNet centroids: the hub arrangement replaced at strength s , every within-cluster offset kept; hits = implant seeds with $z \leq -2$ at $s=0/0.5/1$, $\bar{z}_1$ = mean z at $s=1$, w/b = within/between spread.

model	WordNet-30 frame of record, Gaussian-hub star						Haar-hub star		balanced frame (pre-specified)		
	real z	w/b	hits	s^*	$\bar{z}_1$	tight z at $s=0/0.5/1$	hits $s=0/1$	$\bar{z}_1$	real z	hits	s^*
ViT-T	-1.5	1.5	0/0/0	-	+0.1	-0.6/-1.9/-9.2	0/0	+0.0	-2.1	0/0/1	-
ViT-S	-2.4	1.9	0/0/0	-	-1.2	-0.2/-3.4/-10.1	0/0	-0.7	-1.7	0/0/0	-
ViT-B	-3.6	2.0	0/0/0	-	+0.4	-1.0/-3.9/-11.5	0/0	+0.4	-2.8	0/0/0	-
ViT-L	-4.0	1.9	0/0/0	-	-1.4	-1.6/-4.4/-12.7	0/0	-1.0	-5.6	0/0/0	-
DINO-B	-0.9	2.1	0/0/0	-	-0.3	-1.6/-5.1/-12.1	0/0	-0.3	-1.0	0/0/0	-
DINOv2-S	+0.7	3.0	0/0/0	-	-0.8	-0.2/-3.1/-11.5	0/0	-0.8	+0.1	0/0/0	-
DINOv2-B	-1.2	3.7	0/0/0	-	+0.4	-0.0/-3.1/-11.2	0/0	+0.5	-0.3	0/0/0	-
DINOv2-L	-2.4	3.9	0/0/0	-	+0.2	+0.2/-2.7/-11.4	0/0	+0.2	-1.7	0/0/0	-
DINOv2-G	-1.8	3.9	0/0/0	-	+1.3	+0.2/-3.9/-11.6	0/0	+1.2	-0.9	0/0/0	-
CLIP-B	-2.0	1.3	0/0/0	-	-1.2	-1.7/-3.8/-11.4	0/0	-0.7	-1.3	0/0/0	-
CLIP-L	-0.8	1.5	0/0/3	-	-2.2	-2.4/-5.8/-13.5	0/1	-1.7	-0.7	0/0/0	-
SigLIP-B	-0.9	1.5	0/0/0	-	-1.6	-2.1/-5.0/-11.0	0/0	-1.1	-1.1	0/0/0	-

Table 9: (continued)

(a) Selection diagnostics and mean pairwise ARI at the reference cut.

dataset	configuration	max frac.	CPCC	mean pairwise ARI at the reference cut		
				B/L/G vs block	S, DINO-B vs block	within
ImageNet	euclid-average (deg.)	0.97	0.761	0.03	0.08	0.64
ImageNet	euclid-complete (deg.)	0.83	0.606	0.03	0.17	0.44
ImageNet	euclid-ward	0.48	0.464	0.15	0.38	0.53
ImageNet	cosine-average (selected)	0.40	0.691	0.38	0.35	0.48
ImageNet	cosine-complete	0.32	0.539	0.13	0.17	0.50
ImageNet	cosine-ward	0.37	0.464	0.17	0.39	0.55
CIFAR-100	euclid-average (deg.)	0.80	0.798	0.00	0.41	0.57
CIFAR-100	euclid-complete (deg.)	0.66	0.722	0.17	0.40	0.54
CIFAR-100	euclid-ward	0.33	0.534	0.37	0.65	0.63
CIFAR-100	cosine-average (deg.)	0.53	0.777	0.30	0.54	0.58
CIFAR-100	cosine-complete (selected)	0.39	0.698	0.39	0.51	0.63
CIFAR-100	cosine-ward	0.14	0.584	0.56	0.62	0.61

Table 10: **The categorical island is a chaining artifact of the naive cut; a moderate gap remains under every admissible configuration and closes only in triplet agreement under cosine-average.** (a) Six configurations (metric $\times$ linkage) per dataset with the two quantities that drive the selection criterion of §5.4: the degeneracy diagnostic (largest-cluster fraction at the reference cut, worst model) and the mean cophenetic fidelity (CPCC) of each model’s dendrogram to its own distance matrix; configurations with max-cluster fraction > 0.5 are excluded and, among the rest, the highest CPCC selects the marked row; then the mean pairwise ARI between the cuts at 30 (ImageNet) or 20 (CIFAR-100) clusters. (b) Dendrograms rebuilt from the census cache: ARI at the cut, Pearson correlation between the two trees’ cophenetic distance vectors, and agreement on which pair merges first over 10^4 random class triplets. Under the naive ImageNet configuration, cross-family ARI between the supervised ViTs and the contrastive VLMs reaches 0.82 while every DINOv2-vs-block pair stays at or below 0.06, and the worst model’s largest cluster holds 97% of the classes. Under the selected configuration the DINOv2 family agrees with the block at 0.38 against 0.48 within the block. Under the other 3 admissible ImageNet configurations the DINOv2-vs-block agreement is 0.13–0.17 against 0.50–0.55 within the block; over the admissible configurations and the three measures the median fraction of within-block agreement missing is 0.34, about a third.

(b) Three measures of DINOv2-B/L/G against the supervised+contrastive block, each written as vs block / within block.

dataset	configuration	ARI at the cut	cophenetic corr.	triplet agreement
ImageNet	euclid-average (deg.)	0.03 / 0.64	0.36 / 0.80	0.47 / 0.74
ImageNet	euclid-complete (deg.)	0.03 / 0.44	0.20 / 0.66	0.41 / 0.65
ImageNet	euclid-ward	0.16 / 0.53	0.62 / 0.84	0.67 / 0.85
ImageNet	cosine-average (selected)	0.39 / 0.48	0.48 / 0.78	0.77 / 0.76
ImageNet	cosine-complete	0.13 / 0.49	0.18 / 0.72	0.36 / 0.71
ImageNet	cosine-ward	0.17 / 0.55	0.59 / 0.85	0.69 / 0.86
CIFAR-100	euclid-average (deg.)	0.00 / 0.57	0.15 / 0.78	0.41 / 0.74
CIFAR-100	euclid-complete (deg.)	0.17 / 0.54	0.23 / 0.72	0.41 / 0.66
CIFAR-100	euclid-ward	0.37 / 0.63	0.63 / 0.77	0.70 / 0.79
CIFAR-100	cosine-average (deg.)	0.30 / 0.58	0.42 / 0.72	0.51 / 0.64
CIFAR-100	cosine-complete (selected)	0.39 / 0.63	0.46 / 0.64	0.50 / 0.61
CIFAR-100	cosine-ward	0.56 / 0.61	0.74 / 0.73	0.76 / 0.76

Table 10: (continued)

(a) Alignment with WordNet on ImageNet.

model	Spearman ρ , inter-centroid vs WordNet distance				raw $\hat{\delta}$ of WordNet / random groups
	ρ_{WN}	shuffle	Gaussian	single image	
ViT-T	+0.52	+0.02	+0.01	–	0.146 / 0.159
ViT-S	+0.50	+0.00	–0.00	–	0.110 / 0.130
ViT-B	+0.49	+0.03	+0.01	–	0.092 / 0.120
ViT-L	+0.53	+0.01	+0.03	+0.25	0.104 / 0.117
DINO-B	+0.36	–0.01	–0.01	–	0.084 / 0.117
DINOv2-S	+0.22	+0.01	+0.01	–	0.077 / 0.100
DINOv2-B	+0.18	–0.01	+0.01	–	0.068 / 0.069
DINOv2-L	+0.18	–0.00	–0.03	+0.08	0.060 / 0.058
DINOv2-G	+0.20	–0.03	–0.01	+0.03	0.054 / 0.056
CLIP-B	+0.57	+0.00	+0.01	–	0.126 / 0.174
CLIP-L	+0.59	+0.01	+0.03	+0.26	0.117 / 0.162
SigLIP-B	+0.58	–0.01	+0.01	–	0.118 / 0.158
DeiT-B (IN-1k)	+0.08	+0.01	–	–	–
ViT-B (augreg, IN-1k)	+0.52	+0.01	–	–	–

Table 11: **Alignment with the human taxonomy follows supervision and recipe, the superclasses are recovered by every family, and the recovery is not WordNet circularity.** (a) ρ_{WN} : Spearman correlation between inter-centroid and WordNet tree distances on ImageNet, with the label-shuffle and Gaussian-cloud controls; single image: the same correlation on single-image distances (one image per class); WordNet / random groups: raw $\hat{\delta}$ of WordNet-coherent against random class groupings. (b) ARI of the 20-cluster cut against the true CIFAR-100 superclasses per configuration (degenerate configurations marked; the criterion-selected one is cosine-complete). Paired within each of the 3 admissible configurations, the label-supervised ViT-B recovers more than each DINOv2 model in 11 of 12 comparisons; DINOv2 itself recovers the superclasses at ARI 0.19–0.52, and the best cut over models and configurations reaches 0.61 against zero for permuted labels. Circularity controls: WordNet-coherent groupings read a lower raw $\hat{\delta}$ than random ones in 11 of 12 models, and single-image distances still correlate with WordNet at up to +0.26. The two leaf-label ViT-B/16 of Table 8: the augreg recipe is aligned with WordNet like the IN-21k ViTs (ρ_{WN} +0.52) while DeiT-B is nearly unaligned (+0.08) yet recovers the CIFAR-100 superclasses at ARI up to 0.56, so WordNet alignment is recipe-dependent even among leaf-supervised ViTs.

(b) Recovery of the CIFAR-100 superclasses: ARI of the 20-cluster cut per configuration ($\dagger$: degenerate on CIFAR-100, euclid-average being the naive configuration; $*$: criterion-selected).

model	euclid-avg †	euclid-ward	cosine-avg †	cosine-complete $*$	cosine-ward
ViT-T	0.31	0.38	0.29	0.33	0.33
ViT-S	0.35	0.56	0.37	0.47	0.56
ViT-B	0.39	0.61	0.43	0.45	0.57
ViT-L	0.38	0.55	0.49	0.48	0.60
DINO-B	0.22	0.48	0.28	0.38	0.54
DINOv2-S	0.12	0.51	0.32	0.33	0.49
DINOv2-B	0.02	0.47	0.35	0.42	0.51
DINOv2-L	0.01	0.32	0.25	0.50	0.48
DINOv2-G	0.00	0.19	0.08	0.20	0.52
CLIP-B	0.18	0.48	0.18	0.37	0.48
CLIP-L	0.35	0.52	0.32	0.41	0.54
SigLIP-B	0.22	0.49	0.22	0.37	0.52
DeiT-B (IN-1k)	0.22	0.56	0.23	0.38	0.54
ViT-B (augreg, IN-1k)	0.30	0.44	0.40	0.42	0.48

Table 11: (continued)

(c) DBpedia Classes (219 leaf classes, 3 levels), a text hierarchy independent of WordNet; gains in pp.

model	supremum, Gaussian		census of record			trip.	NC H-R	FS H-R
	$\hat{\delta}_{\max}$	excess	$\hat{\delta}_{99.9}$	excess	$r(p)$			
BGE-base	0.122	-0.028	0.082	-0.021	200 (0.005)	0.89	-0.59	+0.05±0.05
E5-base	0.127	-0.024	0.086	-0.017	200 (0.005)	0.89	-0.43	+0.04±0.03
GTE-base	0.132	-0.020	0.088	-0.017	200 (0.005)	0.89	-0.82	+0.08±0.06

Table 11: (continued) (c) Excess under the original supremum reading (3 Gaussian replicates) and under the census of record (Haar null, 99.9th-percentile statistic, 200 replicates; r = replicates above the real value, left-tail p), sibling-triplet agreement, and the metric gains, marginal as the diagnostic predicts. Tree map on the same classes: no configuration degenerates (largest cluster $\leq 10\%$), the embedders recover the true 70-way level-2 partition at ARI 0.21–0.29 in all 6 configurations and agree with each other at cross-model ARI 0.65–0.82; the configuration the criterion selects there (Euclidean, average linkage) is the one that degenerates on ImageNet, which is why the criterion is applied per dataset. (d) Radial ordering of the four levels under the paper’s projection (chance for strict monotonicity = $1/24 \approx 4.2\%$; shuffle controls ≈ 0), sibling triplet agreement (cosine; chance 0.5), and leaf $\hat{\delta}$ with spectrum-null excess.

(d) HierarCaps (1000 four-level caption chains): a multi-level text hierarchy with no class set.

model	$\rho(\text{level, radius})$	% monotone	trip. L1	trip. L2	$\hat{\delta}$ leaves	excess
BGE-base	+0.80±0.31	49.6	0.88	0.97	0.130	+0.003
E5-base	+0.87±0.19	53.9	0.90	0.97	0.139	+0.010
GTE-base	+0.47±0.55	24.0	0.90	0.97	0.135	+0.006
CLIP-B	+0.64±0.43	27.2	0.86	0.95	0.123	-0.011
CLIP-L	+0.51±0.49	18.7	0.86	0.96	0.117	-0.010
SigLIP-B	+0.32±0.55	9.8	0.83	0.91	0.138	+0.014

Table 11: (continued)

measure	raw	null mean	$\tau_{0.05}$	calibrated	null-centered	pairs $p < 0.05$
mutual-kNN ($k=10$), Euclidean	0.472	0.010	0.015	0.464	0.462	66/66
mutual-kNN, Poincaré	0.425	0.010	0.019	0.414	0.415	66/66
linear CKA, Euclidean	0.633	0.108	0.111	0.577	0.524	66/66
linear CKA, Poincaré	0.640	0.108	0.111	0.586	0.532	66/66

Table 12: **Cross-model agreement survives permutation calibration.** Means over the 66 model pairs of the 12 vision backbones on the 1000 ImageNet class centroids (census cache). Calibration of Gröger et al. (2026) with $K=200$ permutations of the class correspondence and $\alpha=0.05$, scalar (no layer search): $\tau_{0.05}$ is the $\lceil 0.95(K+1) \rceil$ -th order statistic of the observed score and its nulls (eq. 9), $p = (1 + \#\{\text{null} \geq \text{obs}\}) / (K+1)$ (eq. 10), calibrated = $\max\{(\text{obs} - \tau_{0.05}) / (1 - \tau_{0.05}), 0\}$ (eq. 12); “null-centered” is the earlier raw-minus-null-mean variant, kept for continuity. The mKNN null mean equals the analytic chance level $k/(n-1) = 10/999$.

(a) Task accuracies and zero-cost metric gains per cell (10 backbones with cached per-dataset features $\times$ 6 datasets).

model	data	NC acc. (%)		FS acc. (%)		FS advantage over Euclidean (pp)			McNemar
		Eucl.	Poinc.	Eucl.	Poinc.	cosine	radial map	Poinc. _N - cos	NC H vs R
ViT-T	IN	68.1	67.2	97.5	97.6	-0.01	+0.01	+0.08	10^{-22} (-)
	C100	29.3	28.2	62.4	63.1	+1.01	+0.45	+0.55	10^{-5} (-)
	C10	87.5	87.4	87.6	88.1	+0.66	+0.19	+0.29	0.19 (-)
	DTD	63.7	62.5	83.3	83.9	-0.23	+0.32	+0.69	0.03 (-)
	FMNIST	78.5	78.4	81.6	81.9	+0.21	+0.10	+0.31	0.38 (-)
	MNIST	75.9	74.0	74.3	74.6	+0.64	+0.52	+0.20	10^{-13} (-)
ViT-S	IN	77.7	77.5	98.6	98.7	+0.17	-0.02	-0.07	0.09 (-)
	C100	74.9	74.6	92.2	92.4	+0.59	+0.16	-0.00	0.10 (-)
	C10	93.4	93.2	92.0	92.5	+0.56	+0.27	+0.19	0.02 (-)
	DTD	69.3	68.5	85.4	86.0	-0.03	+0.25	+0.73	0.10 (-)
	FMNIST	80.7	80.4	83.7	83.5	+0.28	+0.08	-0.19	0.08 (-)
	MNIST	84.8	84.1	80.5	81.4	+0.34	+0.65	+0.84	10^{-4} (-)
ViT-B	IN	81.6	81.7	98.9	99.0	+0.20	-0.04	-0.05	0.21 (+)
	C100	77.3	77.0	92.5	92.7	+0.56	+0.12	-0.05	0.07 (-)
	C10	94.5	94.0	92.7	93.0	+0.49	+0.18	+0.15	10^{-5} (-)
	DTD	70.8	70.8	85.4	86.2	+0.28	+0.19	+0.69	1.00 (-)
	FMNIST	81.3	81.6	83.4	83.3	+0.03	+0.09	-0.04	0.08 (+)
	MNIST	81.6	80.5	76.8	77.6	+0.19	+0.51	+0.66	10^{-7} (-)
ViT-L	IN	86.1	86.2	99.4	99.4	+0.11	-0.03	-0.02	0.58 (+)
	C100	85.3	85.2	94.8	95.5	+1.17	+0.31	+0.12	0.59 (-)
	C10	97.5	97.5	95.0	95.8	+1.25	+0.35	+0.59	0.76 (+)
	DTD	71.8	71.5	85.6	86.5	+0.13	+0.16	+1.02	0.69 (-)
	FMNIST	83.7	83.5	84.4	84.7	+0.00	+0.17	+0.23	0.12 (-)
	MNIST	84.2	83.6	80.8	81.6	+0.08	+0.42	+0.67	0.00 (-)
DINOv2-S	IN	74.7	74.4	97.4	97.9	+0.68	+0.03	-0.22	0.00 (-)
	C100	73.4	74.4	94.3	95.1	+0.80	+0.15	+0.06	10^{-6} (+)
	C10	95.0	95.2	92.9	93.5	-0.36	+0.13	+1.06	0.14 (+)
	DTD	73.4	73.4	89.8	90.5	+0.30	+0.04	+0.36	1.00 (+)
	FMNIST	82.2	82.2	85.4	85.5	+0.02	+0.07	+0.07	1.00 (-)
	MNIST	83.4	81.9	80.2	80.8	-0.07	+0.45	+0.67	10^{-11} (-)
DINOv2-B	IN	78.9	78.9	97.0	97.8	+1.40	+0.04	-0.63	0.57 (+)
	C100	87.4	87.9	96.5	97.2	+1.11	+0.09	-0.34	10^{-4} (+)
	C10	97.1	97.4	94.7	95.4	+0.27	+0.03	+0.43	10^{-4} (+)
	DTD	74.2	74.7	89.2	90.3	+1.48	+0.01	-0.35	0.31 (+)
	FMNIST	83.2	82.8	85.9	85.9	+0.21	+0.04	-0.18	0.01 (-)
	MNIST	80.6	78.8	76.6	77.1	+0.17	+0.48	+0.47	10^{-16} (-)
DINOv2-L	IN	79.8	80.4	96.2	97.1	+2.13	+0.02	-1.24	10^{-14} (+)
	C100	85.6	87.7	96.1	97.1	+1.80	+0.09	-0.74	10^{-38} (+)
	C10	97.9	98.3	94.1	95.1	+0.92	-0.00	+0.11	10^{-8} (+)
	DTD	74.4	75.2	87.1	88.4	+2.40	+0.03	-0.98	0.11 (+)
	FMNIST	84.4	84.5	87.8	87.9	+0.29	+0.03	-0.03	0.40 (+)
	MNIST	74.1	72.7	70.5	71.1	+0.25	+0.43	+0.46	10^{-12} (-)
DINOv2-G	IN	78.4	78.3	96.2	97.1	+2.57	+0.03	-1.65	0.10 (-)
	C100	89.8	91.0	95.9	97.2	+2.44	+0.13	-1.10	10^{-16} (+)
	C10	97.8	98.2	92.3	93.8	+2.51	-0.02	-1.01	10^{-13} (+)
	DTD	72.9	73.1	87.2	88.5	+2.63	+0.01	-1.38	0.81 (+)
	FMNIST	85.9	85.8	89.0	89.2	+0.38	+0.04	-0.07	0.72 (-)
	MNIST	79.3	78.6	72.9	73.8	+0.30	+0.56	+0.72	0.00 (-)
CLIP-B	IN	61.4	59.9	96.8	96.8	-0.04	-0.00	+0.14	10^{-40} (-)
	C100	65.6	65.1	87.7	88.9	-0.03	+0.44	+1.21	0.07 (-)
	C10	87.9	88.3	89.5	90.1	-0.19	+0.20	+0.90	10^{-4} (+)
	DTD	63.7	63.2	82.2	83.1	-0.23	+0.36	+1.29	0.46 (-)
	FMNIST	75.9	75.1	78.8	78.9	+0.06	+0.17	+0.10	10^{-4} (-)
	MNIST	70.1	66.6	70.9	69.7	+0.04	+0.23	-1.17	10^{-45} (-)
CLIP-L	IN	75.0	74.5	98.4	98.6	+0.01	+0.00	+0.22	10^{-8} (-)
	C100	74.6	74.5	90.0	90.8	+0.12	+0.39	+1.23	0.66 (-)
	C10	95.9	95.9	92.4	93.0	+0.16	+0.25	+0.91	0.59 (-)
	DTD	68.6	69.0	85.2	86.2	-0.08	+0.36	+1.24	0.45 (+)
	FMNIST	76.2	75.9	76.8	77.3	+0.12	+0.33	+0.68	0.09 (-)
	MNIST	85.2	86.1	83.7	84.7	+0.08	+0.58	+1.06	10^{-7} (+)

Table 13: **The calibrated reading predicts the zero-cost gain within datasets, the depth verdict predicts nothing, and cosine is the safe default.** (a) Nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes) accuracy under Euclidean and Poincaré readouts, regenerated from the audited reruns; FS advantage of cosine and of the radial map with Euclidean distances (the map without the metric changes nothing) over Euclidean, and of the Poincaré readout over cosine after L2 normalization; McNemar p for the NC test-set decisions, Poincaré vs Euclidean (order of magnitude when $p < 10^{-3}$; sign of the NC advantage in parentheses).

(c) Best zero-cost metric per backbone: few-shot advantage over Euclidean, mean over the four hierarchical datasets, and which metric collects it (either: within 0.15pp).

model	best-R (pp)	metric
ViT-T	+0.58	either
ViT-S	+0.49	either
ViT-B	+0.51	either
ViT-L	+0.87	either
DINOv2-S	+0.68	H
DINOv2-B	+1.17	cos
DINOv2-L	+1.84	cos
DINOv2-G	+2.54	cos
CLIP-B	+0.67	H
CLIP-L	+0.65	H

Table 13: (continued)

(d) The same correlations on the raw $\hat{\delta}_{99.9}$, the record excess and the depth-test z (Fisher- z CI95; dm: family-demeaned r).

gain	dataset	raw $\hat{\delta}_{99.9}$ r [CI95]	dm	record excess r [CI95]	dm	depth z r [CI95]
NC H-R	ImageNet	-0.84 [-0.96, -0.45]	-0.83	-0.73 [-0.93, -0.19]	-0.61	-0.28 [-0.77, +0.42]
	CIFAR-100	-0.92 [-0.98, -0.68]	-0.58	-0.79 [-0.95, -0.33]	-0.21	+0.52 [-0.16, +0.87]
	CIFAR-10	-0.63 [-0.90, +0.00]	-0.31	-0.67 [-0.91, -0.06]	-0.34	n/a
	DTD	-0.67 [-0.91, -0.07]	-0.67	-0.76 [-0.94, -0.26]	-0.60	n/a
FS H-R	ImageNet	-0.86 [-0.97, -0.50]	-0.56	-0.21 [-0.74, +0.48]	+0.35	+0.27 [-0.43, +0.77]
	CIFAR-100	-0.62 [-0.90, +0.02]	-0.51	-0.79 [-0.95, -0.31]	-0.80	-0.26 [-0.76, +0.45]
	CIFAR-10	-0.78 [-0.95, -0.31]	-0.64	-0.78 [-0.95, -0.30]	-0.66	n/a
	DTD	-0.87 [-0.97, -0.53]	-0.94	-0.98 [-1.00, -0.91]	-0.97	n/a
FS best-R	ImageNet	-0.83 [-0.96, -0.43]	-0.50	-0.08 [-0.68, +0.58]	+0.47	+0.13 [-0.54, +0.70]
	CIFAR-100	-0.77 [-0.94, -0.27]	-0.75	-0.75 [-0.94, -0.23]	-0.65	+0.22 [-0.48, +0.75]
	CIFAR-10	-0.63 [-0.90, -0.01]	-0.55	-0.62 [-0.90, +0.02]	-0.56	n/a
	DTD	-0.81 [-0.95, -0.36]	-0.78	-0.90 [-0.98, -0.63]	-0.84	n/a

Table 13: (continued) (d) The calibrated reading: the raw 99.9th-percentile statistic, the record excess and the depth-test z of Table 8 as predictors. (e) Validation picking retains the oracle advantage and the zero-validation objective rule modestly beats always-cosine. (f) The gate’s value is avoiding the Poincaré projection on flat-label data; cosine is a safe NC default everywhere. (g) The headline $t = 1/\sqrt{2} \approx 0.71$ was fixed a priori and is conservative for FS.

(e) Few-shot metric-selection policies on held-out episodes (500 validation / 500 test per cell).

policy (test advantage over Euclidean, pp)	all 60 cells	hierarchical (40)
always cosine	+0.23	+0.28
objective rule (zero validation)	+0.28	+0.41
δ -gated objective rule	+0.08	+0.13
validation-picked per cell	+0.63	+0.78
test oracle (reference)	+0.64	+0.78

Table 13: (continued)

(f) Nearest-centroid metric-selection policies.

NC policy (advantage over Euclidean, pp)	all 60	flat (20)	hierarchical (40)
always cosine	+0.48	+0.24	+0.59
always Poincaré	-0.24	-0.70	-0.00
gated rule (flat→R)	+0.34	+0.00	+0.51
δ -gated rule	+0.19	+0.00	+0.29

Table 13: (continued)

(g) Sensitivity to the projection target t (36 hierarchical cells).

target t	0.25	0.35	0.41	0.45	0.50	0.58	0.71	0.95
FS adv. mean (pp)	+0.88	+0.87	+0.87	+0.86	+0.84	+0.82	+0.76	+0.56
NC adv. mean (pp)	-0.30	-0.25	-0.23	-0.21	-0.17	-0.12	-0.01	+0.15
% cells FS > 0	97	100	100	100	100	100	100	100

Table 13: (continued)

table	generated file	result file(s)
4	tab_q01_census	expR52_census_haar_p999_200.csv expR54_census_haar_sup_200.csv expR40b_p999census200.csv expR39c_census200_cache.csv expR57_census_cosine_haar_p999_200.csv expR57_text_cosine_haar_p999_200.csv exp10_local_vs_global.csv expR45_convnet_rows.csv exp20_null_ztable.csv
6	tab_q02_text	expR53_text_haar_p999_200.csv expR53_text_haar_sup_200.csv expR53_text_gauss_p999_200.csv expR48b_text_census200_bs1.csv expR57_text_cosine_haar_p999_200.csv exp18_text_nulls.csv exp4_prompt_variation.csv exp17_wordnet_prompts.csv expR49_template_nulls_bs1.csv expR47_extraction_variance.csv
7	tab_q03_sample	expR62_samplelevel_record.csv
8	tab_q04_depth	expR56_depth_variants.csv expR69_depth_haarhubs_summary.csv expR64b_wn30bal_summary.csv expR70_inet1k_supervised.csv expR52_census_haar_p999_200.csv expR63_meru_record.csv expR65_hier_finetune.csv expR71_meru_radii.csv
9	tab_q05_power	expR55_depth_power.csv expR55b_depth_power_leafframe.csv expR64b_wn30.csv expR64b_wn30_summary.csv expR64b_wn30bal_summary.csv expR67_frame_choice.json
10	tab_q06_treemap	expR69_depth_haarhubs_summary.csv exp22_tree_similarity_imagenet.npz exp23_treemap_controls.npz exp23_config_diagnostics.json expR58_treemap_cutfree_summary.csv
11	tab_q07_wordnet	exp3_alignment.csv exp28_recovery_per_config.csv exp8_p1_recovery.csv exp1_delta_controls.csv exp8_p6_pooling.csv expR70_inet1k_supervised.csv exp14_dbpedia.csv expR61_dbpedia_record.csv exp27_dbpedia_treemap.json exp16_hierarcaps.csv
2	tab_q08_robust_final	expR72_budget_record.csv expR72_budget_record_summary.csv expR32_centroid_bootstrap.csv expR59_imagenet_bootstrap_summary.csv expR66_joint_sensitivity_summary.csv expR60_c_sweep_record.csv analysis4_finetuning.csv el_delta_by_layer.csv
13	tab_q09_corollary	table1_regenerated.csv exp2_metric_controls.csv exp2b_normalized_stack.csv exp13_mcnemar.csv night/correlation_cis.csv exp20_null_ztable.csv expR68_corollary_excess.csv exp24_val_metric_selection.csv night/t_sweep.csv
3	tab_q10_calibration	exp1_delta_controls.csv expR42_star_calibration.csv
12	tab_q13_local	exp21b_local_global_K200.csv
5	tab_q14_panel	(model panel: stated in the generator; parameter counts from the model cards)

Table 14: Provenance index: every appendix table is generated by script from the released per-cell result files listed here; no table value is hand-transcribed. The same mapping for main-text tables and figures is annotated in the source and shipped with the code release.